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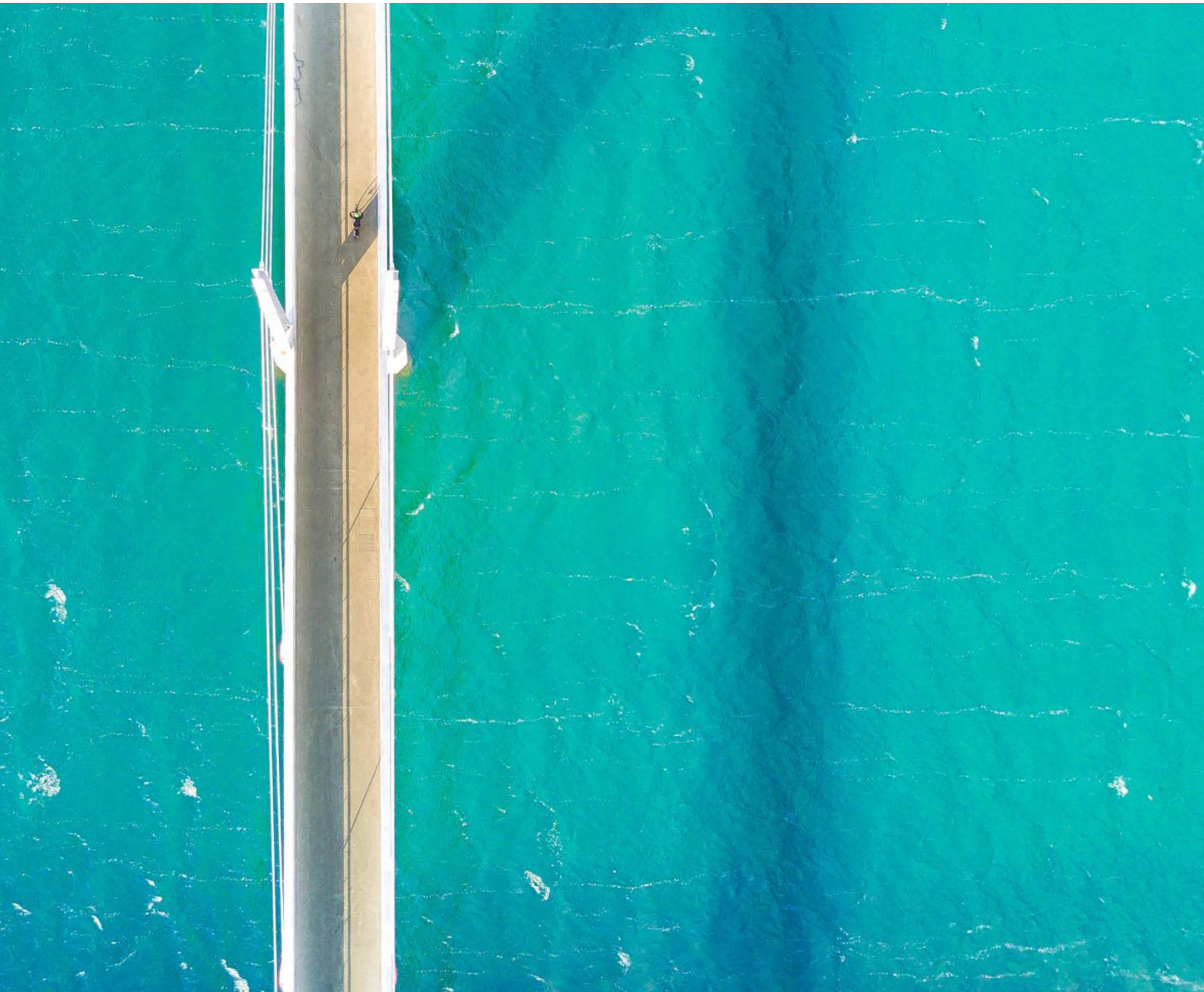
EGYPT | NEW ADMINISTRATIVE CAPITAL PHASE 1 ITS

Concept of Operations for NAC Phase 1 Government Area

Early Deployment

ACUD-ITS-300-RE-001-E
25 Sep 2025



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ABBREVIATIONS

ACUD	Administrative Capital for Urban Development
AD	Active Directory
ARC-IT	Architecture Reference for Cooperative and Intelligent Transportation
ANPR	Automatic Number Plate Recognition
ARLE	Automated Red Light Enforcement
ASE	Automated Speed Enforcement
AST	Agency for State Technology
APTS	Advanced Public Transportation System
ATIS	Advanced Traveler Information System
ATMS	Advanced Traffic Management Systems
CCC	Command & Control Center
COC	City Operations Center
CCTV	Closed Circuit Television
CN	Communication Network
ConOps	Concept of Operations
CFR	Code of Federal Regulations
CVO	Commercial Vehicle Operations
DBOM	Design, Build, Operate, Maintain
DMS	Dynamic Message Sign
DSRC	Dedicated Short Range Communication
EMS	Emergency Management Services
EV	Electric Vehicle
EVCS	Electric Vehicle Charging Station
EVPS	Emergency Vehicle Pre-emption System
FAC	Florida Administrative Code
FMS	Fleet Management System
FTE	Florida's Turnpike Enterprise
GIS	Geographic Information System
GTFS	General Transit Feed Specification
HAR	Highway Advisory Radio
IoT	Internet of Things
ITS	Intelligent Transportation Systems
KPI	Key Performance Indicator
LCS	Lane Control Sign
LPR	License Plate Recognition
MaaS	Mobility-as-a-Service
MoE	Measures of Effectiveness
MOI	Ministry of Interior
MOT	Ministry of Transport
MTBF	Mean Time Between Failures
MTTR	Mean Time To Repair
NAC	New Administrative Capital
NFC	Near-Field Communication
NMV	Non-Motorized Vehicle
NTCIP	National Transportation Communications for Intelligent Transportation Systems Protocol
O&M	Operation & Maintenance
OVDS	Overheight Vehicle Detection System

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RFID	Radio Frequency Identification
RFP	Request for Proposals
RITSA	Regional ITS Architecture
RTVM	Requirements Traceability Verification Matrix
SEA	Systems Engineering Analysis
SIS	Strategic Intermodal System
SLA	Service Level Agreement
SLS	Speed Limit Sign
SNMP	Simple Network Management Protocol
SRD	System Requirement Document
TMC	Traffic Management Center
TPEG	Transport Protocol Experts Group
TSP	Transit Signal Priority
UN	User Needs Identification
USDOT	United States Department of Transport
V&V	Verification & Validation
WIM	Weigh-In Motion

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1 INTRODUCTION

This document presents a Concept of Operations (ConOps) for the Administrative Capital for Urban Development (ACUD) Urban ITS Project / New Administrative Capital (NAC) Government Area Early Deployment, prepared to support the New Administrative Capital's vision of becoming a smart, connected, and efficiently managed city. **A ConOps is a high-level description of a system from the perspective of its stakeholders, intended to capture user needs and operational expectations rather than technical specifications**. It provides a shared understanding of what the system is meant to achieve and how it is expected to be used to be at a high level.

The scope of this ConOps is limited to the NAC Government Area early deployment, serving as a foundation for broader ITS expansion in later phases. The overarching objective is to establish an integrated traffic management capability that enhances safety, improves mobility, and enables coordinated operations across agencies.

1.1 Identification

- **Project Name**

  ACUD Urban Intelligent Transport Systems (ITS) Project – NAC Government Area Early Deployment

- **Position of these ConOps relative to other documentation**

  This ConOps serves as the **high-level conceptual foundation** for the ACUD ITS program. It establishes user needs, operational context, and clarifies stakeholder roles for the NAC Government Area Early Deployment. Future documentation, such as the System Requirement Document (SRD), High Level Design, and Detailed Design Documents will be built on this ConOps. It sits above technical specifications in the hierarchy, aligning stakeholder vision with operational scenarios, and guiding subsequent engineering and procurement phases.

1.2 Purpose And Intended Audience

The purpose of this ConOps document is:

- To describe at a high level the operational concept for the deployment of an ITS within the NAC.
- To communicate user needs, expectations, and system objectives in a clear and structured way, serving as the foundation for subsequent System Requirements Document (SRD), design, and validation.
- To present how the system will enhance traffic efficiency, safety, multimodal integration, and incident management in alignment with national smart city strategies.
- To build a common understanding and consensus among stakeholders on the envisioned ITS functions and operational scenarios.

This ConOps is intended as a strategic and descriptive document, not a detailed technical specification. It sets the operational vision, identifies actors and stakeholders, and outlines how the system is expected to perform under various conditions.

1.3 Document Overview

This Concept of Operations (ConOps) document is structured to provide a clear, logical progression from understanding the existing system to defining the proposed Intelligent Transportation System (ITS) for the NAC. Each section serves a distinct purpose:

Section 1 – Introduction: Establishes the project background, scope, stakeholders, and reference documents, ensuring a common understanding of the ConOps objectives.

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Section 2 – Current System Situation: Describes the present conditions, constraints, and operational environment of the NAC Government Area, highlighting mobility challenges and existing support infrastructure.

Section 3 – Change Justification: Identifies the shortcomings of the current system, justifies the need for improvements, and defines user needs that will guide ITS development.

Section 4 – Concepts for the Proposed System: Presents the vision, objectives, and scope of the new ITS, including operational policies, modes of operation, user interactions, assumptions, constraints, and associated risks.

Section 5 – Operational Scenarios: Illustrates how the ITS will function under normal, incident, planned event, maintenance, and failure conditions, ensuring that stakeholder roles and system responses are clearly understood.

Section 6 – Summary of Impacts: Outlines the expected developmental, organizational, and operational impacts of deploying the ITS.

Section 7 – Analysis of the Proposed System: Evaluates system alternatives, cost and schedule considerations, procurement approaches, and the systems engineering plan, including validation and performance measurement.

Section 8 – Notes: Provides a space for scenario refinements, stakeholder alignment points, and forward-looking integration requirements.

Section 9 – Appendices: Contains supplemental material that supports the main body of the ConOps, including detailed references or extended analyses.

Section 10 – Glossary: Defines key terms and abbreviations to ensure clarity and prevent misinterpretation among readers.

1.4 High Level System Overview

The New Administrative Capital (NAC) of Egypt is located 45 kilometers (28 miles) east Cairo, just outside the second Greater Ring Road and midway to the seaport city of Suez. Conceived as a smart and sustainable city, NAC will serve as the new administrative and financial hub of Egypt, housing government ministries, foreign embassies, and other critical institutions. The city is being developed by the Administrative Capital for Urban Development (ACUD) with the vision of integrating Information & Communication Technology (ICT) to improve quality of life, optimize urban operations, and ensure long term sustainability across economic, social, environmental, and cultural dimensions.

The objective of the ITS project is to establish an intelligent traffic management system that supports the city's broader smart city framework. The system is designed to enable automation and digitization of traffic management, thereby enhancing mobility, safety, efficiency, and citizen experience.

The ConOps provides a high-level description of how the ITS will function within this urban environment, emphasizing integration of ICT and Internet of Things (IoT) technologies to manage traffic flows securely and efficiently. **While ConOps does not present detailed technical specifications, it defines the overarching operational vision, identifies system boundaries, and outlines the service areas** that will guide subsequent system requirements and design phases.

The NAC Phase 1 Government Area Early Deployment ITS project, sponsored by ACUD, represents the initial deployment stage of the broader ITS program. It will be implemented as a pilot in the NAC Phase 1 Government Area and will form the foundation for scaling ITS services citywide. The scope of services for this deployment includes:

1. Advanced Traffic Management System (ATMS)
    - Live traffic surveillance and monitoring.

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- Vehicle monitoring using Radio Frequency Identification (RFID) at intersections, parking gates, and bus detection at stations.
- Traffic incident management.
- Adaptive ramp metering.
- Adaptive traffic signal control (including pedestrian and bicycle crossings).
- Transit Signal Priority (TSP) via RFID.
- Traffic message display, lane closure, diversion & congestion warning via DMS, LCS, SLS.
- Traffic enforcement e.g: speed, red light, Weigh-In Motion (WIM), Overheight Vehicle Detection System (OVDS), Commercial Vehicle Operations (CVO) vehicles.
- Integrated decision support with traffic analysis and prediction, including integration with other ITS platforms.

2. Parking Management Systems
   - Interfaces with the existing COC parking management system.
   - On-street parking space management.
   - Off-street parking management.
3. Public Transportation
   - Transit vehicle tracking.
   - Fleet Management System (FMS) for all transit route operations.
   - Transit security.
4. Data Management
   - COC center data warehouse.
   - CCC center data warehouse.

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1.5 Stakeholders

Stakeholder	Role	Responsibilities
Administrative Capital for Urban Development (ACUD)	Project sponsor, system owner, funding authority	Oversees ITS planning, procurement, and deployment across NAC; ensures integration with smart city vision; coordinates with contractors and consultants.
Ministry of Interior (MOI)	Operational authority for security, enforcement, and traffic law compliance	Operates the Command & Control Center (CCC); manages traffic enforcement (speed, red light, access control); coordinates emergency response.
Ministry of Transport (MOT)	Mobility and transport policy authority	Ensures alignment of NAC ITS with national transport strategy; supports integration of ITS with multimodal public transport and mobility services.
Egis	Technical consultant	Develops ConOps, methodology, and system requirements; engages stakeholders; aligns project with international ITS best practices; supports ACUD in methodology & verification.
Giza Systems	System designers, integrators, and operators (during deployment)	Deliver and integrate ITS subsystems (traffic management, parking, transit, data management); ensure interoperability and compliance with ACUD requirements.
Other Government Ministries / Agencies (Phase 1 Government District Occupants)	Institutional users	Depending on ITS for-staff mobility, parking management, and secure access; coordinate with ACUD for service requirements.
Citizens & Road Users	End users	Utilize ITS-enabled services for mobility, safety, and information; provide feedback for service improvement.

FIGURE 1 – List of Stakeholders

1.6 Referenced Documentation

The development of this Concept of Operations (ConOps) document has drawn upon a range of international standards, project reports, technical references, and supporting documentation. These references provide the technical foundation, methodological consistency, and contextual alignment necessary for ensuring that the proposed concepts are credible, practical, and consistent with both the best global practices and NAC project-specific requirements. The cited materials include internationally recognized engineering standards, prior project deliverables prepared under the NAC ITS initiative, and relevant tools and frameworks adopted by the U.S. Department of Transportation (USDOT).

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The following table lists the references used to develop the concepts presented in this document:

No.	Standards, References & Guidelines
1	ISO/IEC/IEEE 29148 – Systems and Software Engineering Life cycle processes – Requirements Engineering, 2nd Edition, Nov 2018.
2	Egis Report "NAC Phase 1 Architecture Reference for Cooperative and Intelligent Transportation (ARC-IT) Report, Revision C, 02-Mar-2025" (ACUD-ITS-200-RE-001-C).
3	Egis presentation "ACUD Urban ITS Project/ NAC Government Area Early Deployment Mobility Concept Options, Revision B, 23-May-2025" (ACUD-MOB-300-PP-002-B).
4	USDOT ARC-IT SET-IT V9.3 Tool.
5	ACUD ITS RFP – Giza Systems, January 2021.
6	ACUD ITS Addendum, July 2021.
7	Egis Services Methodology for ACUD ITS Project, January 2025.
8	Survey Analysis Report_ACUD-MOB-200-RE-004-B
9	Gov Area Traffic Counts Survey_Report_ACUD-MOB-300-RE-001-B
10	Egis Mobility Concept Options Rev.D
11	ACUD Mobility and ITS Design Project Minutes of Meeting, 2024/2025

FIGURE 2 – Standards, References & Guidelines

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2 CURRENT SYSTEM SITUATION

2.1 Background, Objectives, Transportation Planning & Scope

The New Administrative Capital (NAC) is a flagship project for Egypt, conceived to relieve Cairo's chronic congestion and establish a modern, smart, and sustainable urban center. Phase 1 has already delivered much of the Government Area and residential districts, yet mobility challenges have emerged that affect the city's efficiency and livability.

The NAC Phase 1 **Architecture Reference for Cooperative and Intelligent Transportation (ARC-IT)** frames mobility as a roadmap for integrated transportation across all modes. It emphasizes that successful development depends on reducing congestion, strengthening multimodal connectivity, improving safety for all users, and providing reliable traveler information. This framework ensures that NAC's transport systems evolve as a coordinated, scalable network rather than as isolated projects.

Within this context, the Intelligent Transportation System (ITS) for NAC Phase 1 – Government Area is conceived as a cornerstone of Egypt's transition toward safer, smarter, and more sustainable mobility. Its objectives are not only technical milestones but also part of a broader national and institutional framework that aligns with **Egypt Vision 2030**, the **Administrative Capital for Urban Development (ACUD) smart city strategy**, the **Ministry of Interior's (MOI) safety and enforcement mandate**, and the **ARC-IT transportation planning process** recognized worldwide.

To ensure consistency, clarity, and traceability, the objectives are structured into 4 complementary categories:

1. **General ITS Objectives** – overarching goals that define the system's purpose and long-term value.
2. **ACUD Strategic Objectives** – institutional priorities linking ITS to NAC's role as Egypt's new administrative and smart city hub.
3. **MOI Strategic And Operational Objectives** – safety, enforcement, and security imperatives anchoring the system in law enforcement and public protection.
4. **Transportation Planning Objectives And Strategies (ARC-IT Aligned)** – internationally benchmarked planning goals and performance measures ensuring the NAC ITS is future-ready, interoperable, and aligned with global best practices.

Collectively, these objectives establish the foundation for requirements definition, operational scenarios, validation methods, and performance monitoring across the lifecycle of the NAC Phase 1 ITS.

2.1.1. General ITS Objectives

The ITS for the New Administrative Capital (NAC) Government Area is designed to serve as a transformative mobility system that addresses current challenges while preparing for future demands. The general objectives provide the overarching foundation upon which strategic, institutional, and operational goals are built. These objectives ensure that NAC achieves measurable improvements in safety, efficiency, sustainability, resilience, and user experience.

1. **Safety First**

   Safety is the central pillar of the NAC ITS. The system shall minimize crashes, fatalities, and serious injuries by combining real-time monitoring, automated enforcement, and proactive traffic management. Roadway safety is not limited to motorists but extends to pedestrians, cyclists, and persons with disabilities, who require tailored infrastructure and protective technologies. By embedding continuous safety audits, hazard detection, and rapid emergency response, NAC's ITS will directly support Egypt's national road safety targets and reduce the human and economic costs of accidents.

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2. **Seamless Mobility**

The ITS shall deliver predictable, reliable travel across the Government Area, ensuring that employees, visitors, and logistics operators experience minimal delays. Advanced tools such as adaptive signal control, dynamic rerouting, and coordinated corridor management will be used to optimize traffic flow, particularly during peak commuting hours, special events, and emergencies. The system will also reduce bottlenecks at critical access points, ensuring smooth connectivity to regional corridors and the wider NAC.

3. **Multimodal Integration**

NAC is envisioned as a multi-tiered mobility ecosystem where private vehicles, buses, taxis, freight, shared mobility, electric vehicles (EVs), cyclists, and pedestrians coexist within a single framework. The ITS shall act as the integrator across these modes, supporting unified trip planning, intermodal transfers, and integrated payment systems. For users, this means one mobility experience—whether they are driving, boarding a bus, renting a bike, or charging an EV. This integration reduces dependence on private vehicles and supports long-term sustainability.

4. **Resilience And Continuity of Operations**

As the seat of Egypt's government, the NAC Government Area requires mobility systems that remain operational under all conditions. The ITS shall be designed for resilience, with redundancy in communication, power, and control systems. It shall incorporate disaster recovery plans, failover strategies, and emergency operation modes to guarantee continuity of government functions during incidents, infrastructure failures, or security events. This ensures that critical state activities are not disrupted by transportation breakdowns.

5. **Environmental Sustainability**

The ITS shall actively reduce the environmental footprint of transport activity in NAC. By optimizing traffic flow, reducing idling, and encouraging modal shifts, the system will cut emissions of $PM_{2.5}$, PM_{10}, NO_x, and CO_2, while also lowering fuel consumption. The ITS shall support the deployment of electric vehicle infrastructure, eco-driving applications, and demand management measures that shift trips to cleaner modes. Through these interventions, NAC contributes directly to Egypt's commitments under national climate and environmental strategies.

6. **User EXPERIENCE AND ACCESSIBILITY**

The ITS shall be designed around the end-user experience, ensuring that citizens, government workers, and visitors benefit from reliable and inclusive transport services. This includes real-time traveler information, accessible through roadside displays, mobile applications, and digital platforms, allowing users to make informed choices about routes and modes. Special attention will be given to universal access: infrastructure and services will accommodate people with disabilities, the elderly, and other vulnerable groups, ensuring that the NAC remains an equitable city.

7. **Data-Driven Governance And Decision-Making**

Modern transport systems depend on actionable data. The NAC ITS shall collect, analyze, and share data across agencies to support real-time operations, policy evaluation, and long-term planning. By integrating sensors, cameras, GPS, and vehicle-based systems, NAC will develop a comprehensive mobility data environment. Secure frameworks will be established to ensure data privacy, cyber resilience, and responsible sharing among ACUD, MOI, and national authorities. This enables evidence-based governance and continuous system improvement.

8. **Innovation And Future Readiness**

The NAC ITS is not only about solving today's congestion and safety problems—it is about preparing for tomorrow's mobility landscape. The system shall be designed to integrate emerging technologies such as connected and autonomous vehicles, Mobility-as-a-Service (MaaS) platforms, and AI-enabled traffic management tools. Future scalability is a guiding principle: the ITS platform will evolve with subsequent NAC phases, ensuring interoperability, adaptability, and alignment with global smart city innovations.

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2.1.2. ACUD Strategic Objectives

The Administrative Capital for Urban Development (ACUD), as the master developer of the New Administrative Capital, is responsible for ensuring that the city functions as a world-class smart urban center and the seat of Egypt's government. The ITS is central to this vision, providing the technological backbone that guarantees efficient mobility, safety, and resilience in the Government Area. The following objectives reflect ACUD's priorities for the ITS deployment:

1. **Guarantee Mobility For Government Operations**

   The Government Area is home to ministries, Parliament, the Senate, and the Cabinet. ACUD's first priority is to ensure that government operations are not disrupted by congestion or transport breakdowns. The ITS shall provide secure, reliable, and uninterrupted mobility for public officials, employees, and service deliveries, particularly during peak hours and national events.

2. **Position NAC As A National Smart City Benchmark**

   The NAC is intended to be a flagship for Egypt Vision 2030, demonstrating how technology-driven infrastructure can deliver livability, sustainability, and efficiency. The ITS shall embody this ambition, serving as a model of smart city transport that can be replicated in other Egyptian cities and recognized internationally.

3. **Build Resilience And Scalability Into Infrastructure**

   As the NAC grows beyond Phase 1, the transport network will face increasing demands. ACUD's objective is to establish an ITS framework that is modular, scalable, and adaptable to future phases of city expansion. This includes designing with redundancy, future capacity, and technology upgrades in mind, ensuring that the ITS grows with the city.

4. **Support Sustainable Urban Mobility**

   ACUD emphasizes sustainability across all aspects of NAC development. The ITS shall support green and shared mobility options, including public transport, cycling, walking, and electric vehicles. Demand management strategies, smart parking, and congestion reduction shall be leveraged to reduce reliance on private cars and contribute to lower emissions and healthier air quality.

5. **Integrate With Wider Smart City Services**

   The ITS is not a standalone system but part of NAC's smart services ecosystem, which includes energy, water, security, and governance platforms. ACUD's objective is to ensure full interoperability and data sharing between ITS and other smart city systems, so that transport performance contributes to broader urban efficiency and citizen satisfaction.

6. **Enhance Citizen And Visitor Experience**

   Beyond technical performance, ACUD recognizes that ITS must deliver a positive user experience. This includes clear, reliable traveler information, accessible services for vulnerable groups, and streamlined interactions such as integrated payments and digital platforms. By providing seamless, user-focused services, ITS will enhance the attractiveness of NAC as both a governmental hub and a livable city for residents and visitors.

2.1.3. MOI Strategic And Operational Objectives

The Ministry of Interior (MOI) is entrusted with ensuring safety, security, and law enforcement across Egypt's transport systems. Within the NAC Government Area, MOI's role is particularly critical: it must safeguard the mobility of the nation's administrative core while guaranteeing compliance with traffic laws, rapid emergency response, and the protection of citizens. The ITS shall directly serve MOI's mandate through the following strategic and operational objectives:

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1. **Enhancing Traffic Safety**

   The MOI prioritizes reducing crashes, fatalities, and injuries through real-time monitoring, proactive enforcement, and preventive interventions. The ITS shall integrate automated violation detection, safety audits, and hazard identification tools, ensuring that high-risk areas (e.g., intersections, pedestrian crossings, and work zones) are continuously improved. By embedding vehicle-pedestrian collision prevention systems, safe routes for schools, and real-time hazard alerts, the ITS will serve as a comprehensive safety shield for all users.

2. **Optimizing Traffic Flow And Reducing Congestion**

   Unreliable travel times compromise both efficiency and public safety. The MOI's objective is to use the ITS to minimize congestion and maintain smooth, predictable traffic flow. This will be achieved through adaptive signal control, smart parking management, and advanced simulation tools that anticipate and mitigate bottlenecks. The system shall reduce delays during peak periods and major events while ensuring prioritized movement for emergency and security convoys.

3. **Strengthening Emergency And Incident Response**

   Timely response is vital in protecting lives and securing national functions. The ITS shall enhance MOI's ability to detect incidents instantly, coordinate multi-agency responses, and reduce notification, dispatch, and clearance times. Dedicated priority lanes, vehicle tracking, and integrated communications with Civil Defense and EMS will ensure that emergency vehicles have unobstructed access to critical sites.

4. **Supporting Multimodal Integration And Public Transport**

   The MOI recognizes that safety and efficiency extend beyond private vehicles. The ITS shall help integrate public transport, taxis, bicycles, and pedestrian flows into a cohesive, safe mobility network. By improving public transport reliability, schedule adherence, and passenger information, the ITS supports MOI's broader mission of encouraging modal shift, reducing car dependence, and improving accessibility.

5. **Promoting Sustainability And Environmental Protection**

   The MOI supports Egypt's environmental objectives by reducing transport-related emissions and fuel consumption. ITS measures such as eco-driving support, EV infrastructure integration, and congestion reduction will directly improve air quality and reduce the environmental burden of transport in the Government Area.

6. **Expanding Real-Time Traffic Information And Public Communication**

   A well-informed public is safer and more compliant. The MOI aims to use ITS to provide real-time traveler information—including congestion levels, incidents, and safe routes—through roadside signs, mobile apps, and digital platforms. Two-way communication channels will also strengthen trust between the public and traffic authorities.

7. **Leveraging Data And Analytics For Governance**

   The MOI's enforcement strategy depends on accurate, timely data. The ITS shall enable the collection, storage, and analysis of traffic and enforcement data using sensors, cameras, GPS, and RFID. This will allow authorities to identify bottlenecks, track long-term trends, and guide infrastructure planning. The system shall also support vehicle tracking for public and private fleets, enabling rapid intervention in cases of accidents, emergencies, or criminal activities.

8. **Educating And Engaging The Public**

   Beyond enforcement, the MOI seeks to cultivate a culture of safe and responsible mobility. The ITS shall support awareness campaigns, digital platforms, and in-vehicle systems that educate users about compliance, safe driving behavior, and the benefits of sustainable transport.

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9. **Preparing For Future Mobility Trends**

   The MOI recognizes the coming wave of autonomous vehicles, electric mobility, and Mobility-as-a-Service (MaaS). The ITS shall be designed to accommodate these innovations through supportive infrastructure, updated regulations, and flexible system design, ensuring NAC remains at the forefront of global transport developments.

10. **Ensuring Measurable Outcomes through KPIs**

    The MOI shall evaluate ITS performance against clear Key Performance Indicators (KPIs), including:

    - Uptake and accessibility of public transport.
    - Travel efficiency (journey times, fuel use, reliability).
    - Affordability of services for all users.
    - Safety and security of journeys and facilities.
    - Quality of user experience and satisfaction.
    - Data and personal security, ensuring public trust in ITS systems.

2.1.4. Transportation Planning Objectives And Strategies (ARC-IT Aligned)

The NAC Phase 1 ITS is aligned with the Architecture Reference for Cooperative and Intelligent Transportation (ARC-IT) to ensure that the system is designed within an internationally recognized framework of transportation planning. ARC-IT provides a structured methodology that links user needs, objectives, strategies, and performance measures, ensuring that the NAC ITS evolves as a coordinated, scalable network rather than as a set of isolated projects.

The following objectives and strategies represent the transportation planning perspective of NAC ITS. They are directly traceable to ARC-IT goals and provide measurable outcomes for safety, efficiency, sustainability, and system performance.

1. **Safety**

   Safety remains the most fundamental transportation objective. The NAC ITS shall:

   - Reduce vehicle crashes, fatalities, and severe injuries per vehicle-mile traveled.
   - Target the most common and dangerous crash causes, including red-light violations, speeding, lane departures, weather-related accidents, and secondary crashes.
   - Enhance work zone safety for both workers and road users through dedicated detection, monitoring, and warning systems.
   - Improve protection for vulnerable road users, including pedestrians and cyclists, by integrating intelligent crossings, collision prevention systems, and safe routes to school.

2. **System Efficiency and Reliability**

   An efficient transport system ensures that people and goods move predictably and reliably. The NAC ITS shall:

   - Reduce recurring congestion, measured in terms of daily hours and geographic extent.
   - Improve Travel Time Index (TTI) and Planning Time Index (PTI), ensuring that users can plan trips with confidence.
   - Minimize non-recurring delay caused by incidents, weather events, work zones, and special events.
   - Reduce incident clearance times, directly lowering person-hours of delay.

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- Strengthen traffic signal operations through scheduled retiming, adaptive control, and incident-responsive timing strategies.
- Improve corridor reliability through coordinated management and multimodal balancing.

3. Public Transport and Multimodal Connectivity

Public transport is central to reducing congestion and car dependency in NAC. The ITS shall:

- Increase public transport mode share and improve service reliability.
- Deploy automated fare collection (AFC) and integrated ticketing to simplify travel.
- Enable demand-responsive transport (DRT) and paratransit services to complement fixed-route operations.
- Implement transit signal priority (TSP) to ensure punctual and reliable operations.
- Provide real-time transit information (arrival times, service updates) to passengers via mobile apps and roadside systems.
- Support seamless multimodal trip planning, allowing integration across public transport, walking, cycling, EV use, and shared modes.

4. Traveler Information and Data Sharing

Reliable and accessible information empowers users and agencies alike. The ITS shall:

- Deliver real-time traveler information on congestion, incidents, weather, parking, and multimodal options.
- Disseminate information through Variable Message Signs (VMS), mobile apps, web portals, and social media channels.
- Provide multimodal trip planning tools, covering pre-trip and en-route stages.
- Establish frameworks for data collection, sharing, and interoperability between ACUD, MOI, MOT, and external stakeholders, ensuring coordinated regional planning.

5. Environment and Sustainability

Transportation systems must minimize their environmental footprint. The NAC ITS shall:

- Reduce emissions of $PM_{2.5}$, PM_{10}, NO_x, and CO_2, contributing to cleaner air and healthier communities.
- Lower fuel consumption by smoothing traffic flows and reducing idling.
- Support demand management and parking strategies that discourage single-occupancy vehicle use.
- Promote walking and cycling by integrating non-motorized transport infrastructure into ITS management.
- Enable large-scale adoption of electric vehicles (EVs) by incorporating EV charging station monitoring and management.

6. Freight and Commercial Vehicle Operations

Freight efficiency is vital for economic prosperity. The ITS shall:

- Reduce delays along freight corridors and at intermodal transfer facilities.
- Improve freight travel time reliability, decreasing the number of hours of delay per 1,000 freight vehicle-miles traveled.

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- Enable automated credentialing, permitting, and compliance systems, reducing administrative delays for commercial operators.
- Enhance monitoring of hazardous material transport, ensuring safety and rapid incident response.

7. **Emergency and Incident Management**

   Effective incident management saves lives, reduces delay, and supports resilience. The ITS shall:

   - Reduce average notification, response, and clearance times for incidents.
   - Enable multi-agency coordination between ACUD, MOI, Civil Defense, EMS, and other stakeholders.
   - Support evacuation planning for national emergencies and large-scale events in the Government Area.
   - Incorporate advanced incident detection and response technologies, including video analytics, sensors, and interagency communication platforms.

These transportation planning objectives ensure that the NAC ITS is:

- Fully traceable to ARC-IT goals, strategies, and performance measures.
- Integrated with Egypt Vision 2030 sustainability and mobility targets.
- Scalable and adaptable for future NAC phases, emerging technologies, and long-term governance needs.

By embedding these objectives, the NAC Phase 1 ITS is positioned not only to solve today's challenges but to serve as a national model for coordinated, data-driven, and sustainable transportation planning.

2.2 Operational Constraints (Government Area Focus)

The Government Area of the New Administrative Capital (NAC) is the administrative hub hosting ministries, parliament, and national institutions. While supported by modern infrastructure, the district faces specific operational constraints that affect day-to-day mobility and traffic management. These constraints must be carefully considered in early ITS deployment to ensure efficiency, safety, and sustainability.

Key operational constraints in the Government Area include:

- **Capacity and Congestion:** Traffic surveys i.e. Govt Area Traffic Counts Survey (ACUD-MOB-300-RE-001-B, 2025) indicate recurring peak-hour congestion at major corridors leading to and within the Government District. The high concentration of government employees and service vehicles generates localized bottlenecks, particularly at junctions lacking signalization.
- **Parking Supply and Management:** Field observations (Survey Analysis Report, ACUD-MOB-200-RE-004-B, 2025) show that the current parking supply is insufficient. On-street parking is largely unmanaged, with widespread illegal stopping near ministry complexes and bus layovers, reducing roadway capacity and increasing congestion.
- **Pedestrian Safety and Accessibility:** In several zones, sidewalks are discontinuous or encroached upon by vehicles. Crosswalks are faded or missing, forcing pedestrians to use carriageways, creating safety risks between staff, visitors, and through-traffic. Public Transport Operations: Buses frequently stop in undesignated areas, often blocking lanes or pedestrian zones. This undermines efficiency and safety, and highlights the absence of dedicated curbside management for public transport access.
- **Incident Management:** The Government Area currently lacks centralized monitoring or real-time response capacity. Accidents, breakdowns, or obstructions can cause disproportionate disruption because no ITS-based detection, alerting, or rapid-response systems are in place.

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- **Enforcement and Public Awareness:** Traffic violations such as wrong-way driving, double parking, and unsafe pedestrian crossings are common, but enforcement is limited. The absence of automated enforcement tools and low user awareness of rules diminish compliance.

Summary:

These constraints are **specific to the Government District** and are corroborated by the **NAC Phase 1 ARC-IT Reference (2025)**, the **Survey Analysis Report (ACUD-MOB-200-RE-004-B)**, and the **Gov Area Traffic Counts Survey (ACUD-MOB-300-RE-001-B, 2025)**. Addressing them through targeted ITS deployment—signalization, parking management, enforcement, and real-time monitoring—will be critical to ensuring safe, efficient, and sustainable mobility for Egypt's new seat of government. Description of the current system or situation (Government Area Focus).

2.3 Description of The Current System Or Situation

The Government District in Egypt's New Administrative Capital (NAC) is now fully operational and serves as the central hub for government activity. The area extends over approximately 1.5 million square meters and contains 34 ministries within 10 ministerial complexes, in addition to the Cabinet headquarters, Parliament, and Senate. While many citizen-facing services remain in central Cairo, the NAC has assumed the primary role of hosting the country's administrative functions.

The challenges observed in the Government District are consistent with findings from the NAC Phase 1 ARC-IT Reference (2025), the Survey Analysis Report (ACUD-MOB-200-RE-004-B, 2025), and the Government Area Traffic Counts Survey (ACUD-MOB-300-RE-001-B, 2025). These highlight congestion, parking shortages, pedestrian safety risks, and weak multimodal coordination as critical constraints for early ITS deployment.

Within the district, the current situation presents the following mobility and operational issues:

a) Traffic Congestion & Flow

Congestion is especially visible during peak hours as demand increasingly exceeds available road capacity. The main corridors serving the district experience queuing and slow movement at critical junctions, reflecting both the concentration of ministerial functions in a single area and the limited capacity of supporting infrastructure. This results in reduced accessibility and travel-time reliability for government employees, visitors, and service vehicles.

b) Pedestrian Safety & Accessibility

Sidewalks are not consistently continuous, and in some locations, pedestrians are forced to walk along the carriageway. Marked crosswalks and horizontal markings are either missing, faded, or irregularly applied, creating unsafe crossing behavior. These deficiencies compromise walkability and discourage safe pedestrian movement between government buildings and public transit stops.

c) Public Transport Operations

Public buses often stop in undesignated areas or in the wrong direction and sometimes encroach upon pedestrian paths. This creates safety hazards, disrupts the smooth flow of traffic, and reduces overall efficiency. The absence of properly designed and enforced drop-off and pick-up points further aggravates the situation.

d) Parking & Curbside Management

On-street parking is common but largely unmanaged. Private vehicles and buses occupy unmarked spaces, leading to localized blockages and reduced roadway capacity. The lack of structured regulation for curbside activity undermines traffic efficiency and complicates enforcement.

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e) Public Realm & User Comfort

Several aspects of the pedestrian environment require improvement. Crosswalks are sometimes excessive in number but poorly positioned, walkways near construction zones lack consistent protection, and shaded paths are scarce. These shortcomings diminish pedestrian comfort, especially in hot weather, and reduce the overall attractiveness of walking as a mode of movement within the district. These observations are based on site visits conducted during the assessment phase, supported by field surveys.

f) Incident and Event Management

At present, the Government Area doesn't have ITS-based monitoring or control systems. Incidents such as breakdowns, crashes, or unplanned events cause disproportionate disruption, as there are no mechanisms for automated detection, coordinated response, or traveler information dissemination.

g) Enforcement and User Behavior

Enforcement is carried out manually by on-site police, with limited coverage and inconsistent results. Common violations include wrong-way driving, double parking, and illegal pedestrian crossings, and at uncontrolled junctions. Without automated enforcement or strong awareness campaigns, compliance remains weak.

Summary:

Overall, the Government District is functional but constrained. The combination of congestion, unmanaged parking, pedestrian safety risks, inefficient public transport operations, and weak enforcement reduces efficiency, safety, and comfort. These findings, substantiated by ARC-IT (2025) and ACUD's survey reports, underscore the urgent need for early ITS deployment to establish structured traffic management, coordinated enforcement, and real-time monitoring to support Egypt's new administrative center.

2.4 User Class Profiles

The NAC Phase 1 ConOps involves multiple categories of system users, each with distinct responsibilities in operating, supervising, maintaining, or interacting with the ITS deployment. These roles reflect both day-to-day operations within the NAC and interactions with external stakeholders (such as MOI and ACUD), while ensuring continuity with the wider center-to-center architecture.

- **Administrative Capital for Urban Development (ACUD):** Owner and master developer of NAC; ensures ITS deployment aligns with citywide planning, Vision 2030 objectives, and overall smart city governance. ACUD will also be responsible for city multimodal planning, regulation and governance of intra-city transport operators, and oversight of parking, Electric Vehicle Charging Station (EVCS), Non-Motorized Vehicle (NMV) services, and payment gateways.
- **Ministry of Interior (MOI):** Leads on traffic enforcement, safety, and incident management. MOI is also responsible for traffic management, operating through integration of ITS with traffic police systems, surveillance, and Command & Control Center.
- **Ministry of Transport (MOT):** Responsible for national transport policy, external transport governance, and regulation, ensuring NAC ITS links with national road, rail, and public transport projects.
- **Egis:** Consultant and systems integrator providing technical design, system architecture, and coordination of international best practices with local needs.
- **Giza Systems:** Local technology partner supporting system deployment, maintenance, and localization of ITS solutions to Egyptian regulatory and operational standards.
- **Citizens & Road Users:** End users of ITS services such as real-time traffic info, smart parking, public transport information, and safety applications. Their adoption and feedback will determine long term success.

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In summary, the NAC Phase 1 ConOps assumes that each of these user groups will interact with the ITS according to their defined roles, forming a complete operational ecosystem. The effectiveness of the system depends on clear role definition, reliable system availability, and well-structured coordination between operators, administrators, management, and external stakeholders.

2.5 Support Environment

The government area benefits from existing utility infrastructure that supports ITS roadside equipment (RSE). Both power and fiber optic communication are available, forming a practical foundation for installation and operation. Day-to-day responsibilities are shared between in-house staff and contracted specialists, ensuring that essential services remain functional.

- **Power Supply**: Electrical distribution is already established across the government area, with reliable connections available at multiple points. These existing lines can support ITS equipment without requiring new large-scale upgrades. Routine power maintenance is handled under government service contracts, ensuring continuity of supply.
- **Fiber Optic Communication**: A backbone fiber optic network is in place, with ducts and handholes that can be extended to roadside locations. This network enables high-speed data transfer, which is suitable for ITS applications. Technical interventions such as fiber splicing and testing are managed by external contractors with the necessary expertise.
- **Operational Support**: Government technical staff act as first responders for utility-related issues, coordinating reporting, troubleshooting, and follow-up with vendors. Existing monitoring tools already track power and communication status as part of facility operations, though these are not tailored specifically to ITS.

In summary, the presence of reliable power, a well-established fiber optic backbone, and existing support mechanisms provides a sound basis for ITS connectivity. However, current arrangements remain general-purpose, not dedicated to ITS, which may influence long term efficiency once systems are scaled.

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3 CHANGE JUSTIFICATION

3.1 Justification For Changes

The current state of the Government District demonstrates the need for immediate modernization of traffic management. While basic utilities (power and limited fiber) exist, there are **no traffic signals, controllers, or centralized systems**, and **no roadside equipment** for traffic monitoring or control. As documented in the current situation, day-to-day operations are constrained by congestion at key junctions, gaps in pedestrian infrastructure, unmanaged bus operations, and unregulated curb side activity. These conditions together justify structured ITS deployment.

- **Traffic Congestion & Flow**: Peak-hour queuing and slow movement at critical junctions show that demand is outpacing available capacity and operational control. Without signals or active management, reliability and accessibility for employees, visitors, and services are reduced.
- **Pedestrian Safety & Accessibility**: Discontinuous sidewalks and missing/irregular crosswalks create unsafe crossings and force pedestrians into carriageways. Addressing these risks requires basic control and crossing treatments that do not currently exist.
- **Public Transport Operations**: Undesignated/incorrect bus stopping, wrong-way maneuvers, and encroachment on pedestrian paths disrupt flow and create hazards. Clearly defined and managed curb space cannot be delivered without foundational control and monitoring.
- **Parking & Curbside Management**: Widespread, largely unregulated on-street parking—plus unpredictable stopping by private vehicles and bus queuing in unmarked areas—reduces effective roadway capacity and complicates enforcement.
- **Capability Gap**: The absence of signals, controllers, roadside sensors/cameras, and a traffic management center prevents basic functions such as incident detection, coordinated timing, traveler information, and real-time response.
- **Underused Infrastructure**: Existing power and fiber backbones are not leveraged for operations, representing a missed opportunity to gain immediate performance improvements at relatively low additional civil cost.
- **Strategic Positioning**: Implementing core traffic management and ITS functions is necessary to present the NAC as a modern, well-run capital city and to align operations with the smart-city objectives set for the district.

In summary, change is required because core operational problems are persistent and the tools to manage them are absent. Establishing foundational traffic control and ITS capabilities will address today's congestion and safety issues while putting the NAC on a credible path toward smart, reliable day-to-day operations.

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3.2 User Needs

The success of the Intelligent Transportation System (ITS) for the New Administrative Capital (NAC) depends on its ability to respond to the needs of the key stakeholders who plan, manage, regulate, and use the system every day. **User Needs define what each stakeholder requires to fulfill its mission effectively**, ensuring that the ITS is not a technology-driven platform alone, but a service-driven system that directly supports safety, mobility, and sustainability.

For the NAC Government Area, five primary stakeholder groups shape the system's expectations:

1. **Ministry of Interior (MOI)** – responsible for traffic management, enforcement, road safety, security, and emergency response.
2. **Administrative Capital for Urban Development (ACUD)** – the city developer and operator, accountable for integrated operations, multimodal transportation inside NAC, parking and logistics services, freight movement, and asset management.
3. **Ministry of Transport (MOT)** – the national authority guiding policy, standards, and long-range planning for transportation across Egypt, ensuring NAC integration into the national network.
4. **National Telecommunications Regulatory Authority (NTRA)** – the regulator of communications, data governance, and cybersecurity for ITS platforms.
5. **Travelers and Road Users** – citizens, visitors, and freight operators who rely on the system for safe, reliable, and accessible mobility.

Together, these user needs establish a **stakeholder-driven foundation** for the ITS. They serve as the bridge between national objectives and the functional requirements that will be developed in later stages, ensuring that every technical feature of the NAC ITS can be traced back to a real operational or public need.

User Class	UN ID	User Need Statement	Purpose/Benefit
Ministry of Interior (MOI)	UN-MOI-01	MOI needs to monitor traffic conditions in real time using cameras, sensors, and data feeds.	Provides situational awareness for proactive traffic management.
	UN-MOI-02	MOI needs adaptive traffic signal control and corridor coordination to manage congestion and improve flow.	Reduces delays and supports efficient network operations.
	UN-MOI-03	MOI needs automated detection of traffic violations (speeding, red-light, lane misuse, unauthorized access).	Ensures consistent and scalable enforcement.
	UN-MOI-04	MOI needs violation evidence to be preserved with a secure chain of custody (images, video, metadata).	Guarantees legal defensibility and judicial compliance.
	UN-MOI-05	MOI needs to track and identify vehicles through license plate recognition (LPR), RFID, and GPS inputs.	Strengthens enforcement, security, and investigations.
	UN-MOI-06	MOI needs real-time incident detection for crashes, breakdowns, queues, and abnormal conditions.	Minimizes incident detection time and accelerates response.
	UN-MOI-07	MOI needs automated queue warning and traveller alerts when congestion or hazards occur.	Improves driver awareness and reduces secondary crashes.
	UN-MOI-08	MOI needs to manage work-zone traffic and enforce safety rules in construction areas.	Enhances worker and traveller safety in temporary zones.
	UN-MOI-09	MOI needs to provide emergency vehicle priority and signal pre-emption at intersections.	Shortens response times for law enforcement and EMS.

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User Class	UN ID	User Need Statement	Purpose/Benefit
	UN-MOI-10	MOI needs pre-arrival staging and navigation tools for responders, including unit tracking.	Improves coordination and safety for field teams.
	UN-MOI-11	MOI needs to monitor incident scenes and warn drivers approaching the area.	Protects responders and road users in crash zones.
	UN-MOI-12	MOI needs integrated roadway service patrols to assist motorists, clear debris, and report conditions.	Speeds up clearance and prevents congestion escalation.
	UN-MOI-13	MOI needs coordinated evacuation management, including contraflow, lane-use control, and secure re-entry.	Ensures safe, orderly evacuation during emergencies.
	UN-MOI-14	MOI needs to provide real-time disaster traveller information through DMS, apps, and broadcast channels.	Keeps the public informed and reduces panic in crises.
	UN-MOI-15	MOI needs to monitor and regulate the movement of hazardous materials and dangerous goods.	Minimizes risks from HAZMAT incidents.
	UN-MOI-16	MOI needs to support special-event traffic and security management, including VIP movement coordination.	Maintains safety and order during high-profile events.
	UN-MOI-17	MOI needs secure, interoperable communications with ACUD, MOT, NTRA, and other emergency agencies.	Enables a common operating picture and coordinated action.
	UN-MOI-18	MOI needs cybersecurity protections for all ITS enforcement and safety systems.	Protects integrity and prevents tampering or misuse.
	UN-MOI-19	MOI needs performance dashboards for crash statistics, violation rates, incident durations, and clearance times.	Supports KPI-based monitoring and continuous improvement.
	UN-MOI-20	MOI needs predictive analytics to forecast hotspots, demand surges, and risk conditions.	Enables proactive deployment of enforcement and resources.
	UN-MOI-21	MOI needs integration of data into the Central Command Center (CCC) for oversight and escalation.	Ensures centralized decision-making during critical events.
	UN-MOI-22	MOI needs to prepare for connected and autonomous vehicles by regulating V2X-based enforcement and safety functions.	Future-proofs ITS operations under emerging mobility models.
	UN-MOI-23	MOI needs to manage traffic operations during planned and unplanned events (cultural, political, sports, VIP convoys) through diversions, detours, and corridor control.	Ensures safety, order, and smooth circulation during events.
	UN-MOI-24	MOI needs to provide signal priority for emergency vehicles, enforcement units, transit, and special convoys.	Improves response times and operational efficiency in mixed-use corridors.
	UN-MOI-25	MOI needs to operate and integrate road weather stations (fog, flooding, dust, surface temperature, visibility).	Enhances situational awareness and supports proactive traffic management under adverse weather.
	UN-MOI-26	MOI needs to issue real-time traffic announcements (incidents, congestion, closures, detours, and emergencies) via DMS, apps, broadcast, and web.	Keeps travelers informed and improves compliance with traffic regulations.

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User Class	UN ID	User Need Statement	Purpose/Benefit
	UN-MOI-27	MOI needs to host and manage its enforcement, traffic management, and safety data in the Central Command Center (CCC).	Maintains MOI's operational control, data sovereignty, and legal authority.
	UN-MOI-28	MOI and ACUD need secure, governed data-sharing mechanisms between the CCC and COC, including real-time feeds and interoperability standards.	Enables coordination, common situational awareness, and joint decision-making without compromising data ownership.
Administrative Capital for Urban Development (ACUD)	UN-ACUD-01	ACUD needs a scalable, modular ITS backbone interoperable with smart-city services (utilities, safety, energy, governance).	Ensures ITS integration into NAC's wider smart-city ecosystem.
	UN-ACUD-02	ACUD needs to monitor traffic, parking, and multimodal hub performance through dashboards and KPIs.	Provides real-time insights to optimize urban operations.
	UN-ACUD-03	ACUD needs to manage multimodal traveler information, including park-and-ride, bus, and rail connections.	Enhances accessibility and supports mode shifts.
	UN-ACUD-04	ACUD needs to support transit fare collection and reconciliation across multiple operators.	Ensures seamless payment experiences and financial transparency.
	UN-ACUD-05	ACUD needs to manage freight flows within NAC, including drayage, last-mile delivery, and loading-zone scheduling.	Reduces congestion, improves logistics efficiency.
	UN-ACUD-06	ACUD needs to provide electronic credentialing and roadside safety inspections for freight vehicles in NAC.	Ensures compliance and protects infrastructure.
	UN-ACUD-07	ACUD needs to optimize freight movement through signal priority, dynamic routing, and access to reservation-based parking.	Minimizes bottlenecks and supports just-in-time delivery.
	UN-ACUD-08	ACUD needs to coordinate freight operations with intermodal terminals and logistics providers.	Improves connectivity between NAC and national freight networks.
	UN-ACUD-09	ACUD needs regional parking management, including smart reservations, occupancy monitoring, and dynamic pricing.	Reduces illegal parking and maximizes efficient space use.
	UN-ACUD-10	ACUD needs to manage construction and work-zone activities to reduce disruption.	Maintains traffic flow and safety during city development.
	UN-ACUD-11	ACUD needs roadway environmental monitoring (fog, flooding, sand, temperature) integrated with treatment systems.	Enhances resilience to weather and climate impacts.
	UN-ACUD-12	ACUD needs asset lifecycle management, including preventive maintenance, fault detection, and repair scheduling.	Prolongs ITS asset life and reduces O&M costs.
	UN-ACUD-13	ACUD needs redundancy, failover, and disaster recovery for ITS services.	Ensures continuity of operations under faults or emergencies.
	UN-ACUD-14	ACUD needs secure, interoperable data exchange with MOI, MOT, NTRA, and smart-city command centers.	Provides integrated governance and efficient decision-making.
	UN-ACUD-15	ACUD needs to integrate environmental and sustainability goals into ITS (emission monitoring, eco-driving, EV infrastructure data).	Aligns NAC with Egypt Vision 2030 sustainability objectives.

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User Class	UN ID	User Need Statement	Purpose/Benefit
	UN-ACUD-16	ACUD needs analytics and forecasting tools for demand management, land-use planning, and investment prioritization.	Enables evidence-based planning and scalability.
	UN-ACUD-17	ACUD needs training and knowledge transfer for city operators on ITS tools and operations.	Builds local capacity for long-term system sustainability.
	UN-ACUD-18	ACUD needs to manage smart parking systems, including reservations, dynamic pricing, occupancy monitoring, and integration with traveler information services.	Reduces congestion, supports mode shifts, and maximizes parking utilization.
	UN-ACUD-19	ACUD needs to coordinate parking allocation and logistics support for special events, in alignment with MOI's traffic plans.	Provides orderly access and reduces event-related congestion.
	UN-ACUD-20	ACUD needs to provide and maintain integrated traveler mobile applications offering trip planning, parking reservation, real-time alerts, and payment services.	Enhances user experience, supports multimodal mobility, and increases public adoption of ITS services.
	UN-ACUD-21	ACUD needs to host and manage its multimodal transport, parking, freight, and smart city data in the City Operation Center (COC).	Ensures integrated urban operations and lifecycle asset management.
	UN-ACUD-22	ACUD and MOI need secure, governed data-sharing mechanisms between the CCC and COC, including real-time feeds and interoperability standards.	Enables coordination, common situational awareness, and joint decision-making without compromising data ownership.
Ministry of Transport (MOT)	UN-MOT-01	MOT needs NAC ITS deployments to align with Egypt's national transport master plans and long-range mobility strategies.	Ensures NAC contributes to national mobility goals.
	UN-MOT-02	MOT needs transport performance data (traffic volumes, travel times, modal split, freight flows) from NAC ITS.	Supports demand forecasting, benchmarking, and planning.
	UN-MOT-03	MOT needs NAC systems to be interoperable with national road, rail, and transit networks.	Creates seamless connectivity between NAC and the wider transport system.
	UN-MOT-04	MOT needs to evaluate policies on congestion, safety, emissions, and modal shift using NAC ITS analytics.	Provides evidence for policy refinements and resource allocation.
	UN-MOT-05	MOT needs NAC ITS to support national-level performance monitoring frameworks (e.g., safety KPIs, reliability, freight efficiency).	Enables unified national performance reporting.
	UN-MOT-06	MOT needs integration of NAC ITS with intercity and regional corridors, including freight and passenger movements.	Enhances NAC's role as a national transport hub.
	UN-MOT-07	MOT needs to set technical standards and guidelines for ITS deployments in NAC that can be replicated nationwide.	Promotes interoperability and scalability across Egypt.
	UN-MOT-08	MOT needs multimodal travel demand data (public transport ridership, mode shares, transfer rates).	Informs investment in buses, rail, and active mobility.
	UN-MOT-09	MOT needs NAC ITS outputs to support international benchmarking (Arab, African, global smart city indices).	Positions Egypt as a leader in regional ITS development.

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User Class	UN ID	User Need Statement	Purpose/Benefit
	UN-MOT-10	MOT needs NAC ITS to be integrated with Cairo's transportation systems (Metro, Monorail, LRT, and BRT).	Provides seamless connectivity between Cairo and NAC, reducing reliance on private vehicles.
	UN-MOT-11	MOT needs NAC event traffic and transport management practices to align with national guidelines and be coordinated with intercity transport (rail, metro, monorail, LRT).	Ensures consistency of event mobility strategies across Egypt.
NTRA	UN-NTRA-01	NTRA needs ITS communications to comply with national spectrum allocations and frequency licensing policies.	Ensures lawful, interference-free operation of ITS devices and V2X.
	UN-NTRA-02	NTRA needs resilient, redundant communications networks for ITS centers, field devices, and connected vehicles.	Guarantees reliability of operations under normal and emergency conditions.
	UN-NTRA-03	NTRA needs to enforce cybersecurity protections across ITS data exchanges, platforms, and control centers.	Protects systems against attacks, tampering, and unauthorized access.
	UN-NTRA-04	NTRA needs ITS deployments to adopt secure encryption, authentication, and public key infrastructure (PKI).	Provides trusted communication between agencies and vehicles.
	UN-NTRA-05	NTRA needs data privacy and governance frameworks for ITS (classification, retention, lawful access, anonymization).	Safeguards citizen data and ensures compliance with national law.
	UN-NTRA-06	NTRA needs real-time monitoring and incident response tools for ITS communications networks.	Detects anomalies quickly and mitigates service disruption.
	UN-NTRA-07	NTRA needs NAC ITS deployments to interoperate with national ICT strategies and platforms.	Aligns ITS growth with Egypt's broader digital transformation vision.
	UN-NTRA-08	NTRA needs to coordinate with ACUD, MOI, and MOT to ensure communications standards are unified across agencies.	Supports interoperability and seamless information exchange.
	UN-NTRA-09	NTRA needs NAC ITS systems to be future-ready for 5G+, satellite, and emerging V2X communication technologies.	Ensures long-term scalability and adoption of next-gen mobility.
Travelers and Road Users	UN-TRAV-01	Travelers need accurate, real-time information on traffic conditions, congestion, incidents, and travel times.	Supports informed route choice and reduces delays.
	UN-TRAV-02	Travelers need real-time parking availability and reservation options through multiple channels.	Reduces circulation searching for parking and lowers congestion.
	UN-TRAV-03	Travelers need multimodal trip planning tools integrating private car, public transport, cycling, and walking.	Encourages sustainable choices and seamless door-to-door mobility.
	UN-TRAV-04	Travelers need electronic, contactless payment systems for tolls, parking, and transit.	Increases convenience and reduces barriers to multimodal travel.
	UN-TRAV-05	Travelers need timely and reliable public transport information (schedules, arrivals, disruptions).	Improves confidence in using public transport.
	UN-TRAV-06	Travelers need safety warnings (work-zones, incidents ahead, weather alerts, school zones) delivered through roadside and in-vehicle systems.	Improves driver and pedestrian safety.

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User Class	UN ID	User Need Statement	Purpose/Benefit
	UN-TRAV-07	Travelers need mayday/emergency assistance functions in vehicles or via mobile apps.	Provides rapid help in case of crashes or distress.
	UN-TRAV-08	Travelers need clear evacuation and disaster guidance, including safe routes and re-entry information.	Protects citizens during emergencies.
	UN-TRAV-09	Vulnerable users (pedestrians, cyclists, disabled travelers) need safe crossings, accessible wayfinding, and priority measures.	Ensures inclusive, equitable mobility for all.
	UN-TRAV-10	Commercial drivers need access to freight parking, loading zones, and real-time restrictions updates.	Reduces illegal stops, improves delivery reliability.
	UN-TRAV-11	Commercial drivers need digital compliance and inspection systems that minimize roadside delays.	Improves efficiency while maintaining safety standards.
	UN-TRAV-12	All users need their data to be protected (privacy, consent, security of transactions).	Builds trust in using ITS services.

FIGURE 3 – Key User Group Classification & Needs

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4 CONCEPTS FOR THE PROPOSED SYSTEM

4.1 Background, Objectives, And Scope

The development of the NAC Intelligent Transport System (ITS) responds to the need for a coordinated, technology-driven approach to managing traffic and mobility in Egypt's NAC. Current traffic management practices rely heavily on manual monitoring and response, creating challenges in terms of timeliness, accuracy, and integration across agencies. The proposed ITS is designed to close these gaps by establishing a common operational framework, integrating roadside technologies, and enabling more effective coordination between the Ministry of Interior (MOI), the Ministry of Transport (MOT), ACUD, and other stakeholders.

The objectives outlined in this section are drawn directly from the **ACUD ITS RFP (2021)**, the **NAC Phase 1 ARC-IT Reference (2025)**, and the **Egis Mobility Concept Options (2025)**, which collectively highlight the need for a **centralized ITS platform**. This platform will not only deliver traffic management functions but also support broader smart mobility and city management services.

Objectives

The primary objectives of the NAC ITS include:

- Establishing a **centralized platform for ITS**, which covers not only traffic management but also, broader functions.
- Establishing a **centralized platform** for traffic management and control, integrating all field devices and operational centers.
- Enhancing **incident detection and response**, with clear procedures for coordination between MOI, emergency services, and transport operators.
- Improving **traffic flow reliability** through adaptive signal control and real-time traveler information.
- Providing a **scalable foundation** for future applications such as connected vehicle services, Mobility-as-a-Service (MaaS), and regional system integration.
- Supporting **sustainability** by ensuring interoperability and alignment with national transport and smart city strategies.

Scope

The system concept defined in this ConOps covers:

- Roadside equipment for detection, surveillance, enforcement, and information delivery.
- A communications backbone to support secure and high-speed data exchange between field devices and control centers.
- Integration with the central **Command & Control Center (CCC)** and the **City Operations Center (COC)** for joint incident management, enforcement, and mobility coordination.
- Operational procedures for normal traffic conditions, incident response, planned events, and system disruptions.

Interfaces with external stakeholders such as emergency services, public transport operators, parking and EV charging operators, and municipal authorities. In summary, the proposed NAC ITS will establish a centralized and integrated platform that goes beyond traffic control to encompass smart mobility services, laying the foundation for a scalable and sustainable transport system for Egypt's new capital.

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4.2 Operational Policies & Constraints

This section outlines the operational policies and constraints that will govern the proposed system. These considerations ensure that system operations are consistent, sustainable, and aligned with institutional capabilities, noting that no prior ITS-specific operational work has been developed for ACUD.

General Operational Policies

- **Hours of Operation:** Define whether the system is expected to operate on a 24/7 basis, or with scheduled service windows, considering traffic demand and staffing requirements.
- **Staffing Requirements:** Identify the operational roles needed to manage the system (e.g., control center operators, maintenance crews, incident response teams). Policies should specify minimum staffing levels during normal and peak operations.
- **Space & Facilities:** Outline the operational space requirements such as control centers, equipment rooms, or on-site field facilities.
- **Hardware & Software Constraints:** Note constraints relating to interoperability with existing systems, minimum hardware performance, and requirements for secure and reliable software platforms.

Reference To Prior Work

Where applicable, operational policies and constraints already defined in earlier ConOps, feasibility studies, or planning documents will be referenced rather than repeated here. This ensures continuity between the current ConOps and previously approved operational frameworks. At present, no prior ITS-specific operational policies or constraints have been developed for ACUD; this ConOps therefore provides the initial reference framework for future operational governance.

Business Rules & Constraints

Operational policies will be supplemented by system-specific business rules, including but not limited to:

- Rules for priority use (e.g., emergency vehicle access, planned events).
- Traffic management rules under system downtime or degraded conditions.
- Contingency procedures for failures in sensors, telecommunications, or central software platforms.
- Configuration and mode-change rules during system upgrades, expansion, or transition periods.

System Limitations & Constraints

Physical and operational limitations that may affect service delivery include:

- **Infrastructure Constraints:** Geometric limitations of existing corridors, location of utilities, and integration with current roadway infrastructure.
- **Financial Constraints:** Limitations in available O&M funding, particularly where toll revenues or government allocations may fluctuate.
- **Operational Transition:** Procedures required during the commissioning or expansion of facilities, when temporary reductions in service levels may occur.
- **Technology Dependencies:** Reliance on third-party communication networks, power supply stability, and resilience against cyber or physical disruptions.

Supporting Documentation

Relevant studies, operational manuals, or corridor-specific policies will be listed and referenced here, such Relevant studies and frameworks that inform operational policies and constraints include:

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- Request for Proposals (RFP) for the NAC ITS project, which establishes baseline requirements and operational expectations.
- NAC Phase 1 Architecture Reference for Cooperative and Intelligent Transportation (ARC-IT).
- National and regional ITS standards and operating procedures (e.g., NTCIP, GTFS, TPEG, SNMP).
- ACUD transportation and feasibility studies addressing traffic demand, congestion, and integration with Cairo's networks.
- Technical memoranda and operational manuals on incident response, maintenance scheduling, and data management (where available, or to be developed as part of implementation).

4.3 Description of The Proposed System

The proposed ITS deployment for the New Administrative Capital (NAC) introduces new subsystems and the proposed ITS deployment for the NAC introduces subsystems and capabilities aligned with Regional Architecture service packages, expanding the role of the existing Control & Command Center (CCC) through its integrated Traffic Management Center (TMC). Together, these enhancements establish NAC's first-generation smart mobility platform by connecting roadside systems, enforcement tools, and traveler information services under a cohesive framework.

Planned changes include:

- **Adaptive Traffic Control System (ATCS) Service Package:** Deployment of fully signalized intersections with adaptive control, replacing the current non-signalized baseline, enabling real-time demand management at key corridors and junctions.
- **Automated Traffic Enforcement (ATE) Service Packages:** Deployment of standard Automated Red Light Enforcement (ARLE), Automated Speed Enforcement (ASE), and Automatic Number Plate Recognition (ANPR) along with License Plate Recognition (LPR) systems providing NAC with its first automated enforcement framework.
- **Advanced Traffic Management System (ATMS) Surveillance & Incident Management Service Package:** Installation of CCTV cameras and automated incident detection to support proactive monitoring and rapid response through the TMC.
- **Advanced Public Transportation System (APTS) Service Packages:** Introduction of bus fleet tracking, schedule adherence monitoring, and passenger information systems, enabling operators to oversee service quality and inform passengers in real time.
- **Advanced Traveler Information Services (ATIS) Service Packages:** Deployment of Dynamic Message Signs (DMS) and digital channels to deliver real-time traffic, incident, and public transport updates to road users.
- **Emergency Vehicle Pre-emption System (EVPS) Service Package:** Provision of priority at intersections for ambulances, fire, and police vehicles, improving emergency response times.
- **ITS Data Platform Active Directory (AD) Service Packages:** Establishment of a unified integration layer that consolidates inputs from all subsystems into dashboards, KPI, and reports, supporting ACUD's strategic mobility management.
- **Communication Network (CN) Expansion Support Service Packages:** Extension of ACUD's communications backbone to connect field devices with the TMC, ensuring secure, reliable, and high-availability links.

In summary, the proposed system transitions NAC from a baseline of non-signalized roads and minimal roadside technology into a fully integrated ITS environment. Built on Regional Architecture service packages, the system equips ACUD and its TMC with modular, scalable, and future-ready applications for smart mobility management.

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4.4 Modes of Operation

The NAC ITS will operate under a set of defined modes to ensure reliable, safe, and efficient management of traffic, enforcement, and public transport. These modes reflect the complementary operational requirements of both the **ACUD Control and Operations Center (COC)** and the **MOI-led Control and Command Center (CCC)**, and their coordinated integration with new ITS subsystems.

- The **COC (ACUD)** serves as the primary day-to-day operator of ITS functions, including traffic monitoring, public transport tracking, parking management, and traveler information.
- The **CCC (MOI)** provides higher level oversight across safety, security, and law enforcement domains, and has authority to assume control during emergencies, special events, or incidents requiring inter-agency coordination.

Planned modes of operation include:

a) Standard / Normal Operations

- The COC manages day-to-day ITS operations, including traffic signals, CCTV monitoring, enforcement systems, public transport tracking, and traveler information dissemination.
- The CCC maintains situational oversight, monitors critical incidents, and ensures integration with broader city and national security functions.

b) After-Hours Operations

- Reduced staffing at the COC, supported by automated monitoring and system alerts.
- Critical incidents are automatically escalated to the CCC & on-call emergency staff for immediate action.

c) Maintenance Mode

- COC staff coordinate with field technicians to take subsystems (signals, CCTV, DMS, enforcement cameras) offline for scheduled maintenance.
- The system generates maintenance logs and alerts, ensuring operational KPI are not disrupted.
- The CCC is informed of any downtime that may affect enforcement or security monitoring.

d) Diagnostic / Testing Mode

- The COC isolates subsystems for calibration, device testing, or integration of new components.
- Testing is performed in a controlled mode to avoid disruption of live traffic or enforcement functions.
- CCC visibility is maintained to ensure security and enforcement are unaffected.

e) Emergency Operations

- The CCC assumes command to coordinate police, fire, ambulance, and civil protection responses.
- The COC provides operational support, including traffic signal preemption, dynamic traveler information updates, and coordination of road clearance.
- Direct linkage between both centers ensures rapid incident detection, response, and inter-agency collaboration.

f) Fail-Safe / Backup Mode:

- Network redundancy rings in the event of communications loss, device malfunction, or COC downtime, predefined fail-safe mechanisms are triggered (e.g., default signal timings, offline camera recording).

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- The COC restores operations and synchronizes data once connectivity resumes.
- The CCC is kept informed to maintain situational awareness and security continuity.

g) Training Mode (Simulated Environment):

- The COC uses a controlled simulation environment to train operators on normal, emergency, and contingency scenarios without impacting live operations.
- The CCC also conducts periodic joint training for incident command and emergency procedures, ensuring interoperability between both centers.

4.5 User Involvement And Interaction

The proposed ITS for the NAC will serve a range of user groups, each with distinct roles and levels of interaction. Since the system will be newly deployed across the city, user interaction represents a significant change compared to the current situation, where most functions are handled through conventional or manual processes.

User Groups and Roles:

- **Control and Operations Center (COC) Operators** – Monitor real-time traffic and transport conditions through the ITS platform, respond to incidents, and coordinate with field units.
- **Control and Command Center (CCC) Staff** – Provide high level oversight, ensuring strategic coordination across multiple domains (security, utilities, transport), and integrate ITS operations into broader city management.
- **Traffic Police and Enforcement Officers** – Access system data for incident management, violation detection, and enforcement actions.
- **Maintenance Technicians** – Use diagnostic and monitoring tools to track equipment health, perform preventive maintenance, and respond to technical failures.
- **Emergency Management Services (EMS) Ambulance, Civil Defense** – Receive priority notifications, routing support, and direct coordination through the ITS platform to optimize emergency response times.
- **Public Transport Operators** – Use system-generated information for fleet tracking, service performance, and passenger information dissemination.
- **Parking Operators** – Manage smart parking assets, integrate availability and payment systems, and coordinate enforcement with MOI patrols.
- **Non-Motorized Vehicle (NMV) Operators** – Oversee micromobility services (bicycles, scooters), monitor docking stations, and manage user payments.
- **Electric Vehicle Charging Station (EVCS) Operators** – Monitor charging infrastructure, ensure uptime, and manage digital payment transactions.
- **System Administrators** – Manage access control, user accounts, and system configuration to ensure smooth operations.

Methods of Interaction

- **System Dashboards and Workstations** – Operators in the COC will interact through user-friendly dashboards displaying traffic, incidents, and sensor feeds.
- **Mobile and Handheld Devices** – Field staff (e.g., police, emergency responders) will access data via secure mobile applications.

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- **Automated Alerts and Reports** – Stakeholders will receive scheduled and event-driven notifications, ensuring proactive decision-making.
- **Public-Facing Interfaces** – Travelers will interact indirectly through dynamic message signs (DMS), public transport information displays, and mobile apps providing travel and route guidance.

Purposes of Interaction

- **Active Control** – Adjusting system parameters (e.g., traffic routing, incident response protocols) based on real-time conditions.
- **Monitoring** – Continuous observation of network performance, safety, and operations.
- **Incident Management** – Coordinated interaction between operators, enforcement, and emergency services for rapid incident resolution.
- **System Maintenance** – Diagnostic tools for equipment status checks, fault reporting, and preventive interventions.

Interactions Among Users

- Operators in the COC will coordinate with CCC staff for escalations and multi-agency alignment.
- Police, transport operators, and emergency services will interact with each other through shared incident response workflows embedded in the system.
- Maintenance staff will interact with operators to schedule or respond to system downtimes.
- Parking, NMV, and EVCS operators will interact with both ACUD and MOI to ensure seamless integration of user-facing services, compliance with enforcement protocols, and alignment with broader mobility objectives.

In summary, the system establishes new layers of user interaction not present today, transitioning from manual and fragmented practices to an integrated, technology-driven model that now includes parking, NMV, EV charging operators, and field traffic patrols as key stakeholders.

4.6 Assumptions & Constraints

The following assumptions and constraints apply to the proposed ITS enhancements under this project.

Assumptions

- The **Command & Control Center (CCC)** and **City Operations Center (COC)** are already operational and will remain the primary hubs for system management.
- Existing fiber and communications backbone will provide sufficient capacity for integration of new ITS devices.
- Field devices to be deployed (CCTV, DMS, detection) will comply with international ITS standards and be interoperable with the existing platforms.
- System architecture will align with Egyptian national ITS strategy and international best practices.
- No major changes to institutional responsibilities are anticipated; existing operator structures will continue to manage daily operations.
- Data centers will provide the necessary hosting environment for ITS applications, and sufficient data warehousing capacity will be available to support storage, analytics, and reporting needs.

Constraints

- Budgetary and procurement limitations may restrict the scale or phasing of ITS deployment.

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- Geometric and right-of-way limitations may affect placement of gantries, cabinets, or field equipment.
- Policy and regulatory constraints may apply to data sharing, particularly with respect to video recording, personal data, or inter-agency information exchange.
- Staffing levels at CCC/COC may constrain the extent of 24/7 active monitoring unless additional resources are committed.
- Construction work will need to minimize impacts on live traffic flow, requiring phased implementation and coordination with traffic management authorities.
- Interoperability with existing subsystems may require custom integration or adaptation, as some legacy systems have limited upgrade paths.
- Environmental factors such as high temperatures, dust, and humidity may constrain equipment performance and necessitate more frequent maintenance.

4.7 Risks

The ACUD project introduces a range of operational, technical, and organizational risks. While the overall system architecture is designed for resilience, several risk factors require consideration during planning, deployment, and operations. A preliminary risk register is provided below.

Risk #	Risk Owner	Description of Risk & Impact	Likelihood (1–4)	Impact (1–4)	Mitigation Strategy
1	ACUD	Integration risk: complexity of interfacing ITS subsystems with the central platform may delay implementation or reduce functionality.	3	3	Early integration testing, phased rollouts, and clear interface specifications.
2	MOT / ACUD	Operational readiness: lack of trained operators or delayed staffing for the control center could affect service continuity at launch.	2	4	Develop and implement a training program in advance; phased staffing and shadow operations.
3	MOT / ACUD	Integration with MOT-operated transport modes (e.g., buses, metro) may face interoperability or governance challenges, causing delays in service harmonization.	3	4	Establish formal data-sharing agreements, adopt common interface standards, and conduct joint integration workshops with MOT and other TMC.
4	Technology Vendor	Cybersecurity vulnerabilities: risk of intrusion or data breaches due to centralized ITS data flows.	2	4	Apply national cybersecurity standards, conduct penetration testing, & establish monitoring protocols.

FIGURE 4 – Risk Register, Impacts & Strategy

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4.8 Support Environment

The support environment for the ACUD ITS deployment will build upon existing operational and administrative capacities within the **Command & Control Center (CCC)** and the **City Operations Center (COC)**, while incorporating new processes and resources specific to the proposed systems. The support environment will ensure system reliability, maintainability, and lifecycle continuity.

a) Support Agencies and Facilities

The CCC and COC will remain the primary operational entities, with additional roles defined for maintenance teams, IT support groups, and vendor-provided technical support. A centralized operations and maintenance facility will serve as the hub for equipment storage, repairs, and spares management.

b) Equipment and Software Maintenance

Preventive and corrective maintenance schedules will be developed for all ITS equipment. Support software will include asset management tools, diagnostic applications, and system monitoring platforms. Software updates, patches, and configuration management will be performed under controlled release cycles to maintain interoperability and minimize downtime.

c) Repair and Replacement Criteria

Faulty devices will be subjected to field level troubleshooting. If repair is not viable within the agreed service level timelines, replacement with pre-procured spare units before Mean Time Between Failures (MTBF) will occur. A defined Mean Time To Repair (MTTR) and replacement standards will be established as part of the maintenance strategy.

d) Maintenance Cycles and Levels

- **Level 1**: Routine inspections and cleaning by field crews.
- **Level 2**: Corrective maintenance for identified faults, including software resets and component swaps.
- **Level 3**: Advanced diagnostics and repairs carried out at the central facility or by vendor-certified technicians.

e) Storage, Distribution, and Supply Methods

Spare equipment and consumables will be stored in climate-controlled facilities near the CCC/COC. Distribution to field sites will be managed through a logistics framework that supports rapid deployment and minimizes downtime.

f) Future Agreements and Contracts

Long term maintenance contracts will be established with technology vendors for specialized systems. Service Level Agreement (SLA) will define uptime targets, response times, and escalation procedures.

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5 OPERATIONAL SCENARIOS

The operational scenarios outlined in this section provide a practical high level view of how the planned ITS systems will function once deployed within the NAC. These scenarios illustrate expected system behavior across a range of real-world conditions, highlighting the interactions between the Ministry of Interior (MOI) Command & Control Center (CCC), the NAC City Operations Center (COC), field systems, and end users.

By examining normal day-to-day operations, responses to incidents, and recovery from failures, the scenarios demonstrate how the ITS will enhance safety, mobility, and overall efficiency of transport management. This structured approach ensures that the envisioned system is not only technically sound but also operationally ready to meet the needs of Egypt's stakeholders in the future mobility environment.

5.1 Normal Operations Scenarios

Normal operations within the NAC illustrate how ITS services function during day-to-day traffic, parking, and traveler activities. The following subsections provide detailed narrative scenarios, demonstrating how enforcement, parking, public transport, non-motorized vehicle (NMV) services, and electric vehicle (EV) charging operate in practice. Each scenario highlights how ACUD (via the City Operations Center), the Ministry of Interior (via the Command & Control Center), the Ministry of Transport, and the traveling public interact with the ITS platform to deliver seamless, reliable mobility.

■ Traffic Enforcement Scenario

A vehicle exceeds the posted speed limit along a major NAC corridor. Roadside enforcement cameras automatically capture the violation and transmit the data to both the COC (ACUD) and the CCC (MOI). CCC operators verify the violation by cross-checking camera footage and sensor data, reducing the risk of false alarms. Once confirmed, the CCC generates an electronic citation that is digitally linked to the driver's license plate and account. If the violation is severe or repeated, a patrol vehicle is dispatched to intercept the offender on the roadway.

From the user's perspective, the driver receives a notification through SMS and the NAC mobility app, which contains clear details of the violation and instructions for secure, cashless payment. The fine is settled digitally, and receipts are stored in the driver's account. Meanwhile, ACUD logs the event in its system dashboards, monitoring patterns of recurring violations across locations. MOT receives aggregated enforcement data to analyze trends and feed into national road safety policy, such as reviewing speed limits, signal placement, or corridor-specific interventions. This cycle ensures that enforcement is automated, transparent, and seamlessly integrated with both user-facing and policy-making functions.

■ Parking Management & Enforcement Scenario

A commuter heading to the NAC downtown checks the NAC mobility app before starting their trip. The app displays real-time availability of both on-street and off-street parking spaces, with predictive analytics highlighting areas likely to fill up soon. The driver reserves a space in an underground garage, guaranteeing availability on arrival.

At the entry point, license plate recognition verifies the reservation, lifts the gate, and triggers automatic payment through the driver's linked wallet. Occupancy data is instantly shared with the COC (ACUD), ensuring continuous visibility of parking demand, and with the CCC (MOI) for enforcement monitoring.

If the commuter overstays the paid duration or parks outside designated zones, enforcement is triggered automatically. Fixed sensors detect violations, and the CCC dispatches parking enforcement patrols or issues electronic citations. This approach reduces the need for manual policing and improves compliance.

From a broader system perspective, ACUD aggregates parking data to optimize allocation, while MOT integrates this data into national-level demand management strategies. Users benefit from reduced circling

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for spaces, simplified payments, and transparent enforcement, creating a smoother downtown travel experience.

Public Transport Scenario

A resident plans their morning commute using the NAC traveler app. The app integrates real-time bus schedules, vehicle locations, and expected delays, allowing the rider to choose the fastest option. Upon boarding, the passenger pays using a mobility card or mobile wallet, with the fare transaction processed through MOT's national transport platform.

As the trip progresses, the COC (ACUD) monitors bus adherence to schedules and identifies delays caused by congestion. When service reliability is threatened, the CCC (MOI) provides traffic signal priority along congested intersections, ensuring buses move more efficiently. MOT coordinates directly with operators to reroute buses or deploy standby vehicles if demand spikes unexpectedly.

For passengers, the experience is dynamic and transparent. Push notifications, digital signage at stops, and in-vehicle displays provide continuous updates on travel times, delays, and alternative connections (such as shuttle services or future metro links). The integration of transport modes within one app reinforces the NAC's vision of seamless multimodal mobility, while MOT benefits from accurate ridership data to inform long-term planning.

NMV Monitoring & Payment Scenario

A student chooses to rent a shared bicycle to reach their university campus. Using the NAC mobility app, they locate the nearest docking station, scan a QR code to unlock the bicycle, and complete digital payment instantly. The transaction is linked to the user's mobility account, where usage history and costs are stored.

The COC (ACUD) continuously monitors availability across docking stations, ensuring that bikes and scooters are evenly distributed. When some stations become depleted or overloaded, rebalancing crews are dispatched based on real-time data. If a user abandons a bike outside designated docking zones, the system flags the violation, and the CCC (MOI) enforces penalties or dispatches field staff to retrieve the vehicle.

From a user perspective, NMV trips are fully integrated into the wider transport ecosystem: the app highlights connections to bus routes or parking hubs, allowing riders to combine cycling with other modes. This supports affordable, eco-friendly travel while ensuring that operations remain orderly and compliant. MOT incorporates NMV data into its multimodal integration strategy, furthering sustainable transport policies across Egypt.

EV Station Monitoring & Payment Scenario

A visitor driving an electric vehicle (EV) uses the NAC mobility app to identify an available charging bay. The app shows real-time availability, allows advance reservations, and provides pricing information. On arrival, the EV is authenticated through either NFC card tap or license plate recognition, which authorizes charging and initiates the billing process.

During charging, the driver can monitor progress via the app, which sends a notification once charging is complete. Payment is processed automatically through the linked wallet, and a digital receipt is issued. The COC (ACUD) monitors station uptime and immediately dispatches maintenance teams if a charger malfunctions.

Meanwhile, the CCC (MOI) enforces fair usage by preventing overstays in charging bays, ensuring turnover and availability for other drivers. MOT collects aggregated usage data to inform Egypt's broader EV adoption strategy, including policies on charging infrastructure deployment and integration with renewable energy sources. From the user's perspective, EV charging in the NAC is reliable, predictable, and fully integrated with other mobility services.

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Cross-Agency Roles and System Functions in Normal Operations

While the above scenarios illustrate specific operational flows, the underlying responsibilities remain consistent across daily functions:

- **ACUD (via COC):** Ensures system availability, manages smart parking, NMV, and EV platforms, and provides integrated dashboards.
- **MOI (via CCC):** Leads enforcement, traffic control, and curbside regulation.
- **MOT:** Oversees public transport integration, ensures compliance with national standards, and links NAC ITS with national platforms.
- **Public Users:** Access multimodal options, benefit from optimized signals, and engage with digital services for parking, EV charging, NMV, and journey planning.

System-wide functions include continuous monitoring of traffic and parking, adaptive control of signals, multimodal traveler information, integrated payments, and logging of operational data for predictive analysis. These functions ensure that the NAC's ITS platform operates seamlessly during normal conditions, forming the baseline for the more complex responses outlined in Sections 5.2 through 5.5.

5.2 Incident Response Scenarios

When a traffic incident occurs in the NAC area, the ITS provides tools for rapid detection, coordinated management, and efficient recovery. The scenario shows how ACUD, the Ministry of Interior (MOI), and the Ministry of Transport (MOT) work together to minimize disruption while ensuring public safety and mobility.

a) Incident Detection and Verification

- The ITS automatically flags abnormal events such as sudden speed drops, stopped vehicles, debris on the roadway, or multi-vehicle collisions using CCTV analytics, roadside sensors, and automatic incident detection software.
- Simultaneous alerts are generated for both the COC (ACUD) and the CCC (MOI Traffic Police), ensuring that both operational and enforcement perspectives are aligned from the start.
- CCC operators verify the incident through live camera feeds, cross-checking from multiple sensors and, when available, connected vehicle data. This reduces false alarms and confirms critical details like the number of lanes affected, the severity of injuries, and environmental risks (e.g., fuel spills).
- Early verification allows the CCC to immediately classify the incident type (minor, major, hazardous) and trigger predefined response protocols.

b) Traffic Management and Response Coordination

- MOI Traffic Police at the CCC assume command, dispatching patrol vehicles, ambulances, and fire brigades to the scene within minutes. Integration with emergency service CAD (computer-aided dispatch) ensures accurate location information and priority routing.
- Traffic signal systems near the incident are automatically adjusted: signals are extended to clear traffic queues, emergency vehicles are granted green-light priority, and diversion routes are activated.
- ACUD ensures continuous availability of ITS platforms — communication networks, parking databases, and Geographic Information System (GIS) so the CCC always has an up-to-date situational picture. The COC also provides staging or holding areas (e.g., parking facilities) for emergency response teams and tow trucks.
- MOT ensures that broader mobility implications are addressed: public buses may be rerouted, coordination with regional highways is activated, and national freight movements are monitored to avoid cascading congestion outside the NAC.

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c) Public Communication and Guidance

- Road users approaching the incident zone are warned via roadside DMS, which display lane closures, recommended detours, and estimated clearance times.
- Mobile applications push targeted alerts, while in-vehicle navigation systems receive live updates through integrated data feeds, ensuring rerouting advice reaches drivers before they encounter congestion.
- Public transport riders are notified through apps and station displays if bus lines or shuttle services are delayed or detoured.
- Information is consistently provided in Arabic and English, tailored to NAC's mixed population of residents, commuters, and visitors.

d) Post-Incident Recovery

- Once the scene is cleared, CCC operators gradually restore normal signal operations, deactivate diversions, and return the system to baseline traffic conditions.
- ACUD's platforms automatically compile an incident report: timeline of detection, verification, dispatch, clearance time, and overall traffic impact. These reports feed into ACUD's operational dashboards and are shared with MOT for broader performance analysis.
- MOT reviews the incident data in the context of national transport policy, identifying whether systemic improvements are needed (e.g., faster detection algorithms, better coordination with ambulance services, or enhanced bus rerouting procedures).
- Lessons learned are incorporated into both NAC's operating procedures and Egypt's wider smart mobility framework.

Summary

a) **ACUD (COC):** Provides system availability, ensures unified ITS infrastructure performance, and supports response logistics.

b) **MOI (CCC):** Leads incident command, field response, and traffic management.

c) **MOT:** Oversees integration with national networks and aligns incident handling with strategic mobility goals.

d) **Public:** Receives accurate, real-time guidance, benefiting from reduced delays, safer travel, and faster recovery.

5.3 Planned Events Scenarios

Large-scale planned events in the NAC area such as diplomatic summits, cultural festivals, sporting competitions, or large concerts create predictable surges in travel demand. These events require early coordination, seamless traffic and crowd management, real-time user communication, and data-driven recovery. ACUD, the Ministry of Interior (MOI), and the Ministry of Transport (MOT) play complementary roles, supported by the ITS platform.

■ Pre-Event Planning and Coordination

a) Event organizers provide detailed schedules, venue layouts, and estimated attendance figures to the MOI's Command & Control Center (CCC). This ensures that both security and transport planning are embedded from the outset.

b) ACUD integrates these inputs into its ITS modeling tools to forecast expected demand, simulate network impacts, and identify likely congestion hotspots (e.g., venue approach roads, key intersections, and parking areas).

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c) Parking demand is projected across multiple facilities, with pre-designated lots for VIPs, event staff, and the general public. Shuttle staging zones and park-and-ride areas are identified in advance.

d) MOT aligns multimodal transport services by scheduling additional bus trips, ensuring BRT headways are shortened, and adjusting rail connections. Intercity services are reinforced to handle the surge of visitors arriving from outside the NAC.

e) A joint operations plan is agreed upon between MOI, ACUD, and MOT, defining responsibilities for venue access control, traffic diversions, and public transport support.

Traffic and Crowd Management During the Event

a) MOI, operating from the CCC, activates special event traffic management protocols:

- Establishing restricted access zones for VIPs, delegations, and secure areas.
- Assigning priority routes for emergency services and key dignitaries, with automated signal preemption along their corridors.
- Adjusting signal timings to handle peak pedestrian crossings near venues and dispersing traffic more effectively across the grid.
- Deploying additional traffic police to enforce temporary regulations and respond rapidly to on-the-ground incidents.

b) ACUD systems provide continuous operational support by:

- Dynamically updating parking availability and guiding vehicles to the nearest open lots.
- Publishing live congestion maps and incident alerts across operator dashboards.
- Using CCTV, drones, and sensor data to monitor both traffic and pedestrian movement in and around event sites.

c) MOT ensures that shuttle services run at high frequency, coordinates real-time information with transit operators, and manages standby fleets for sudden demand surges.

User Communication and Public Guidance

a) Travelers receive real-time updates on road closures, shuttle availability, and parking guidance via roadside DMS, official event apps, and NAC mobility platforms.

b) Push notifications through mobile apps and in-vehicle navigation provide route alternatives, estimated travel times, and instructions for safe pedestrian movement.

c) Targeted communication is delivered:

- **Residents** are directed to alternate routes to avoid congestion.
- **VIP delegations** receive secure, private route guidance integrated into police escort planning.
- **Pedestrians** are guided with wayfinding apps and temporary signage that show walking paths, shuttle stops, and entry gates.

d) Public messaging is bilingual (Arabic/English) and synchronized across platforms, ensuring consistency and reducing confusion.

Post-Event Recovery

a) Once the event concludes, MOI phases out traffic restrictions in stages, reopening access roads gradually while maintaining a police presence until traffic stabilizes.

b) ACUD systems automatically capture performance data, including parking utilization, travel times, congestion levels, and compliance with traffic management measures.

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c) This data is compiled into after-action reports, highlighting network bottlenecks, incident hotspots, and overall system resilience.

d) MOT evaluates how public transport absorbed demand, measuring ridership spikes, shuttle reliability, and intercity connectivity. Lessons learned are incorporated into operational playbooks for future events.

Summary

a) **ACUD (COC):** Provides predictive modeling, real-time ITS support, and post-event analytics.

b) **MOI (CCC):** Leads operational control, enforcement, and VIP/secure access management.

c) **MOT:** Adjusts multimodal transport capacity, ensuring integrated service delivery and resilience.

d) **Public:** Benefits from clear information, safer travel, minimized congestion, and smooth access to and from major events.

5.4 Maintenance Scenarios

Ensuring the long term reliability of the NAC ITS requires structured maintenance activities that balance **technical upkeep, operational continuity, and public safety**. ACUD leads the technical side of asset management, while the MOI—through the Command & Control Center (CCC)—ensures traffic safety and coordination during maintenance activities.

a) Planned Preventive Maintenance

- ACUD develops a rolling maintenance schedule covering traffic signals, CCTV cameras, environmental sensors, roadside units, communications infrastructure, and central software platforms.
- Maintenance windows are strategically chosen (e.g., overnight or low-traffic periods) to minimize disruption.
- Prior to each activity, ACUD provides the CCC with detailed notifications on expected duration, asset location, and any possible traffic impacts.
- MOI operators, from the CCC, adjust traffic signal timings, patrol deployments, or lane management plans to maintain smooth flow while works are underway.
- Preventive maintenance is not limited to inspections but also includes software patching, firmware upgrades, and calibration of field devices to maintain system accuracy.

b) Corrective Maintenance and Fault Response

- Faults are detected either through automated self-diagnostics (e.g., camera offline alerts, sensor calibration errors) or through operator reports in the CCC.
- Each fault is logged in the integrated incident management system, which assigns priority levels based on severity (e.g., critical failure affecting safety vs. minor sensor misalignment).
- ACUD dispatches specialized technical teams to the field. For major failures (e.g., a dark signal at a critical junction), response crews are given priority routes and immediate access.
- While repairs are in progress, MOI staff manage temporary interventions: deploying officers to direct traffic, adjusting nearby signal cycles, or issuing automated detours through the ITS.

c) Traffic and Public Safety Measures

- For works requiring partial closures, lane reductions, or temporary service interruptions, the MOI coordinates diversions and enforcement directly from the CCC.
- Dynamic Message Sign (DMS), in-app notifications, and integrations with third-party navigation platforms (e.g., Google Maps, Waze) push real-time updates to road users.

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- Travelers are informed of alternate routes, expected delays, and safety advisories, reducing the risk of secondary incidents near work zones.
- For high visibility works, MOI may deploy roadside patrol units for added safety and rapid response to unexpected traffic build-up.

d) Performance Monitoring and Continuous Improvement

- Once maintenance is completed, ACUD runs system diagnostics and functionality tests to confirm restored performance.
- A formal "close-out" report is generated, shared with MOI operators in the CCC, and logged into the asset management system.
- Historical fault and maintenance data is analyzed to identify recurring issues, enabling predictive maintenance strategies and guiding future technology upgrades.
- Joint debriefs between ACUD and MOI ensure lessons learned are applied, strengthening coordination protocols and minimizing recurrence of high-impact faults.

Summary

a) **ACUD (COC):** Plans and executes preventive/corrective maintenance, ensures technical restoration, and manages asset performance records.

b) **MOI (CCC):** Maintains traffic safety, manages diversions and interim measures, and provides operational oversight during work.

c) **Public:** Receives timely updates, alternate routes, and minimized disruption thanks to coordinated maintenance management.

5.5 Failure Scenarios

Even with robust design, redundancy, and preventive maintenance, ITS components in the NAC may experience partial or complete failures due to technical malfunctions, cyber disruptions, power outages, or physical damage. The response framework ensures traffic safety, mobility continuity, and rapid system recovery until full service is restored.

■ Detection & Alerting

a) Failures are identified automatically through system health diagnostics, which continuously monitor device status, communication links, and server performance.

b) Alerts are triggered when abnormal conditions occur (e.g., dark traffic signals, frozen camera feeds, sensor malfunctions, or network downtime).

c) Operators at the CCC/COC may also manually flag issues observed during real-time monitoring.

d) Each incident is logged into the incident management system with details on severity, location, and potential operational impacts.

■ Immediate MOI Control Measures

a) The MOI, from the CCC, initiates contingency traffic management protocols to maintain safety and order.

b) Depending on the failure type, measures may include:

- Deploying officers to critical intersections or corridors where signal control has failed.
- Switching affected junctions to pre-timed signal plans as an interim fallback.
- Broadcasting urgent safety messages and detour guidance via DMS, radio, and emergency channels.

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- Temporarily adjusting lane-use strategies or restricting access to high-risk areas until systems are stabilized.

c) These actions safeguard public safety while buying time for technical teams to restore system function.

ACUD Technical Response

a) ACUD dispatches specialized technical teams to diagnose and repair the failed subsystem, whether it involves roadside equipment (signal controllers, detectors, CCTV), backend systems (servers, databases), or communication infrastructure (fiber, wireless links).

b) Field interventions are carefully coordinated with MOI staff to ensure work zones remain safe, particularly in live traffic environments.

c) If backup systems (e.g., redundant servers or parallel communication lines) exist, ACUD activates them immediately to limit operational downtime.

d) Critical repairs are prioritized based on traffic impact, with escalation procedures in place for major outages affecting large parts of the network.

Public Communication and Guidance

a) For short disruptions, MOI manages traffic locally without requiring public alerts.

b) For prolonged outages, road users are informed in real time through:

- Dynamic Message Signs (DMS) with rerouting or hazard warnings.
- Mobile and navigation apps are integrated with the ITS platform (e.g., live detours, congestion updates).
- Social media and official event channels when wide public awareness is necessary.

c) This keeps drivers informed, supports compliance with temporary traffic measures, and reduces secondary congestion effects.

System Recovery and Validation

a) Once the fault is repaired, ACUD gradually restores system functionality in coordination with MOI operators in the CCC.

b) Validation checks are performed, including field verification of signals, communication tests, and system-wide diagnostics.

c) The CCC confirms that normal operational standards are met before declaring the system fully restored.

d) A **joint post-incident review** between ACUD and MOI identifies the root cause, evaluates the timeliness of responses, and recommends preventive actions (e.g., upgrades, redundancy improvements, or revised procedures).

Summary

a) **MOI (CCC):** Ensures immediate traffic safety through contingency plans, officer deployment, and traveler communication.

b) **ACUD (COC):** Leads technical fault diagnosis, repair, and restoration of ITS assets.

c) **CCC:** Serves as the hub for coordination, logging, oversight, and ensuring a smooth return to full service.

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6 SUMMARY OF IMPACTS

The deployment of the proposed ITS system will generate a range of impacts that affect stakeholders, operators, and roadway users. These impacts can be categorized into three main types: developmental impacts, organizational impacts, and operational impacts.

Developmental Impacts

During the implementation phase, temporary disruptions may occur due to system installation, integration with existing infrastructure, and testing activities. This includes the deployment of roadside equipment (sensors, cameras, communication units), installation of control software at the City Operations Center (COC), and configuration of interfaces with other systems. These activities may require scheduled lane closures, adjustments to ongoing operations, and coordination with contractors and utility providers.

Organizational Impacts

The new system introduces enhanced data sharing, automated monitoring, and coordinated incident response capabilities. Stakeholders will need to adapt their workflows to make use of the new tools, such as real-time dashboards, automated alerts, and integrated communication platforms. Training will be required for COC operators, field maintenance staff, and partner stakeholders to ensure consistent use of the new functions. Clear governance procedures will be essential to manage responsibilities across multiple entities and to ensure interoperability with existing systems.

Operational Impacts

Once operational, the system will improve situational awareness, incident detection, and response efficiency. However, it will also place new demands on COC staff, who will be required to manage larger volumes of real-time data and coordinate responses across multiple platforms. Additional maintenance requirements will arise for roadside ITS equipment, communication networks, and central control software. These operational impacts will require ongoing resource allocation, both in terms of staffing and budget, to ensure that system reliability and performance standards are sustained.

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7 ANALYSIS OF THE PROPOSED SYSTEM

7.1 Alternatives

In developing the proposed ITS architecture and operational concept for NAC Phase 1, several alternatives were evaluated to balance cost, complexity, and long term scalability. The focus was on how ITS functions such as traffic management, incident response, and mobility services would be coordinated between MOI and ACUD.

Nothing Alternative – No Coordinated ITS Deployment

a) **Description:** Agencies continue operating with traditional, non-integrated methods. Traffic management relies on manual officer deployment, static signal plans, and isolated systems with little to no real-time data exchange.

b) **Advantages:** No upfront capital cost; avoids institutional restructuring or integration complexity.

c) **Disadvantages:**

- Severe limitations in situational awareness, with no consolidated view of traffic, incidents, or mobility services.
- High reliance on manual responses, leading to delays, inefficiency, and increased crash risks.
- Inability to provide real-time traveler information or adaptive management, resulting in congestion and public dissatisfaction.
- Contradicts NAC's positioning as a "smart capital," undermining its long term mobility and sustainability objectives.

d) **Assessment:** Not viable. Failing to deploy an integrated ITS would result in unsafe, inefficient, and fragmented operations inconsistent with the smart city mandate.

Alternative 1 – Decentralized Operations by Individual Agencies

a) **Description**: Each agency (ACUD for infrastructure operations, MOI for enforcement and traffic, service providers for transport/parking) would manage ITS functions independently with only minimal data exchange.

b) **Advantages**: Lower initial integration cost; agencies retain autonomy and existing systems.

c) **Disadvantages**: Fragmented operations, limited situational awareness, delayed response times, inconsistent user experience for the public.

d) **Assessment**: Not suitable for NAC's vision of a smart, fully integrated city.

Alternative 2 – Full Centralized Control by MOI through the CCC

a) **Description**: MOI operates and controls all ITS functions through the Command & Control Center (CCC), including traffic management, incident response, and mobility services, while ACUD's role is limited to infrastructure support.

b) **Advantages**: Clear chain of command and single point of control; strong law enforcement integration for safety and security.

c) **Disadvantages**: Reduces ACUD's role in managing civil operations such as parking, utilities, and facility-related mobility services; risks overburdening MOI with responsibilities beyond its core mandate.

d) **Assessment**: Effective for traffic enforcement and security, but limits broader ITS integration and flexibility.

Alternative 3 – Shared, Integrated Operations (Hybrid Model)

a) **Description**: MOI operates the CCC as the central command for traffic management and enforcement, while ACUD manages complementary mobility and support services such as parking, asset management,

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and maintenance coordination. Data and operations are integrated through a common ITS platform with seamless data exchange.

b) **Advantages**: Balances institutional mandates; supports efficient incident response with MOI leading through CCC and ACUD providing asset and service visibility; scalable to future ITS applications such as CAVs and MaaS.

c) **Disadvantages**: Requires robust interagency coordination and governance; higher initial integration cost compared to standalone systems.

d) **Assessment**: Preferred approach because it aligns with smart city principles, maintains MOI authority over safety and enforcement, and enables ACUD to optimize supporting operations, ensuring an integrated, user-centric mobility experience.

Rationale For The Selected Alternative

The hybrid integrated model (Alternative 3) was selected because it best balances operational efficiency, stakeholder responsibilities, and future scalability. It ensures MOI retains authority and leadership through the CCC, while ACUD contributes to system optimization and city-wide service integration. This structure delivers seamless operations for the public and supports NAC's long term smart mobility vision.

7.2 Cost, Schedule, And Procurement Options

The implementation of the proposed ITS and COC enhancements involves significant investment, requiring careful consideration of cost, development schedule, and procurement strategy.

- **Estimated Costs**

  The project is expected to incur costs across system design, hardware and software acquisition, system integration, acceptance testing, and training. In addition, annual operations and maintenance costs will apply, particularly for systems such as surveillance, traffic management, and communications. Costs are presented at an order-of-magnitude level, subject to refinement during detailed design.

- **Schedule**

  A phased implementation schedule is anticipated, beginning with procurement and system design, followed by staged deployment of priority subsystems (traffic management, incident response, and parking integration), and culminating in full operational readiness. Overall deployment is estimated to take approximately 18–24 months following contract award.

- **Procurement Options**

  Procurement may follow different models depending on ACUD's and MOI's strategic priorities. Options include direct public procurement by ACUD or MOI for system components, a Design, Build, Operate, Maintain (DBOM) model, or long term service contracts with specialized ITS operators. Consideration will be given to integration with existing platforms and ensuring clear lines of responsibility for system ownership and operations.

In summary, the cost, schedule, and procurement planning provide a framework for advancing the COC and ITS systems in a structured, financially sustainable, and timely manner, aligned with stakeholder roles and expectations.

7.3 Systems Engineering Plan

The ConOps provides the foundation for the remaining systems engineering process, guiding design, integration, and deployment of the COC and ITS systems. The following elements will be central to this process:

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- **Requirements Traceability and Verification Matrix (RTVM):** The RTVM will be derived from the user needs identified in Section 3.2. This matrix will link each user need to specific system requirements, design elements, and test cases, ensuring traceability and verification at all stages of development.
- **Systems Verification & Validation (V&V) Plan:** A structured verification process will confirm that the system, subsystems, and interfaces meet defined requirements. The plan will outline test methods, acceptance criteria, and responsibilities, ensuring that delivered components conform to technical specifications.
- **System Design Process:** Building on Section 4, "Concepts for the Proposed System," the design process will refine system architecture, interfaces, and functional specifications. Emphasis will be placed on interoperability, scalability, and alignment with stakeholder operational requirements.
- **Systems Validation Plan:** Validation activities will confirm that the implemented system satisfies user needs and performance measures, as identified in Section 3.2, and that operational scenarios described in Section 5 are effectively supported. Validation will include stakeholder reviews, demonstrations, and field evaluations.

In summary, the systems engineering plan ensures that the ConOps is carried forward into disciplined technical processes, linking user needs to system design, verification, and validation to achieve a fully functional and operationally effective system.

7.4 Performance Measurement for System Validation

A System Validation will be conducted to determine whether the NAC Phase 1 system meets the user needs and project goals identified in Section 3.2. Unlike verification, which tests equipment and system functions against specifications, validation measures the overall effectiveness of the deployed system in improving operations and outcomes for end users.

Validation Approach

System validation will follow the FHWA systems engineering "V-Model," which links user needs defined early in the project to validation activities at deployment. Planning for validation is embedded in the Requirements Traceability Verification Matrix (RTVM), ensuring that each user need is directly associated with a validation measure. The approach combines before–after evaluation, operational testing, and stakeholder feedback to demonstrate whether NAC Phase 1 is achieving its intended outcomes.

- **Baseline and After Conditions** – Baseline data will be collected where feasible, using existing sources (e.g., incident logs, traffic counts, transit performance reports), supplemented by temporary collection (e.g., speed detectors, CCTV analytics) if needed. **After data will be collected during live system operations once NAC Phase 1 functions are in place.**
- **Operational Field Testing** – Validation will emphasize **end-to-end system performance under real-world conditions**, including incident response, traveler information dissemination, and transit operations during normal, peak, and incident scenarios.
- **Stakeholder Involvement** – Structured feedback from operators, emergency services, and partner agencies will be incorporated to confirm usability and operational effectiveness, complementing the quantitative data.
- **Measures of Effectiveness (MoEs)** – A defined set of MoEs (listed below) will provide measurable evidence of system performance and benefits.
- **Reporting** – Findings will be documented in a System Validation Report, confirming whether user needs and project goals have been met and identifying any gaps for consideration in future phases.

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Measures of Effectiveness (MoE)

The following MoE will guide validation:

- Incident Detection and Response – Reduction in average incident detection time, verification time, and clearance time.
- Travel Time Reliability – Improvement in corridor level travel time consistency during peak and non-peak conditions.
- Traffic Flow Efficiency – Reduction in recurring congestion and delay on key approaches and within the NAC area.
- Information Accuracy and Timeliness – Percentage of traveler information messages disseminated within target timeframes and verified for accuracy.
- System Availability – Percentage uptime of core NAC systems, including data exchange and control center operations.
- Stakeholder Satisfaction – Feedback from operators, emergency services, and other agencies on usability and effectiveness of NAC functions.

Alignment with Regional Architecture Objectives

In addition, validation will assess the system's alignment with the Regional Architecture objectives and performance measures defined for NAC Phase 1, including:

- Enhancing transportation planning and performance measurement with operational data from all modes.
- Using ITS to improve access to alternative modes of transportation and to reduce emissions and vehicle miles traveled.
- Increasing the percentage of transportation agencies that integrate operational data and use NAC systems as a management resource.
- Improving the efficiency of the surface transportation system.
- Enhancing bus service levels through higher frequencies, longer hours, and expanded route networks.
- Improving operational efficiency to maximize use of existing fleets.
- Upgrading facilities such as bus stops, shelters, and terminals.
- Improving the travel time reliability of public transit.
- Monitoring and managing bus occupancy to avoid overcrowding.
- Reducing roadway congestion, both recurring and related to special events.
- Developing applications and payment portals to enhance traveler trust and experience.
  Providing real-time traveler information and multimodal planning tools.
- Increasing the number of trips taken by green modes such as transit, walking, and cycling.
- Collecting, archiving, and analyzing operational transportation data to inform future investment planning.
- Expanding use of ridesharing and non-motorized modes by improving information on their availability and access.
- Improving safety for walking and bicycling through ITS-enabled measures.
- Enhancing bicycle and pedestrian safety through new facilities incorporated into transportation infrastructure.

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Data Collection And Analysis

To support before–after analysis, baseline data will be collected where feasible, supplemented with temporary data collection (e.g., speed detectors, CCTV analytics) if needed. After data will be collected during system operation once NAC Phase 1 functions are in place. These requirements will be reflected in the RTVM to ensure that performance measurement is planned from the outset.

The validation plan will expand these measures further, defining data sources, collection intervals, and analytical methods. This ensures that validation is not treated as an afterthought, but as an integral part of NAC system implementation planning.

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8 NOTES

1. **Scenario Refinement**

   The operational scenarios described in this ConOps are indicative and will be refined during detailed design. Updates to assumptions—such as communication latency, redundancy provisions, cybersecurity measures, and coordination with emergency services—will be recorded in this section as they are confirmed.

2. **Stakeholder Alignment**

   Clarifications or agreements between stakeholders will be logged here to maintain alignment across agencies. This may include responsibilities, escalation procedures, and data-sharing arrangements. Keeping these points in one place helps avoid misinterpretation during later phases.

3. **Change Tracking**

   Revisions to assumptions, roles, or system requirements will be noted with the date and responsible party. This provides a clear record for project reviews and ensures all parties are working with the most up-to-date information.

4. **Future Integration**

   As the NAC ITS evolves, this section will also capture forward-looking requirements, such as integration with new technologies or national systems, so that operational scenarios remain adaptable.

5. **Post-Business Analysis Updates**

   Following the completion of business analysis and change management activities, this ConOps document shall be revised to reflect updated requirements, stakeholder inputs, and operational considerations.

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9 APPENDICES

There are no Appendices at the moment.

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